



**NEW HORIZON
COLLEGE OF ENGINEERING**

Autonomous College Permanently Affiliated to VTU, Approved by AICTE & UGC
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A PROJECT REPORT

for

Mini Project – 2025-2026 (24CSE48)

on

PHARMACY MANAGEMENT SYSTEM

Submitted by

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CERTIFICATE

This is to certify that the Mini Project (24CSE48) work titled “**PHARMACY MANAGEMENT SYSTEM**” is a bonafide work carried out by **H SURUTHI(1NH24CS220) and TAMMINEEDI LAKSHMI CHINMAYEE (1NH24CS221)** in partial fulfillment of the degree of **Bachelor of Engineering** in **COMPUTER SCIENCE AND ENGINEERING** of the New Horizon College of Engineering during the academic year **2025-26**.

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ABSTRACT

The Pharmacy Management System is a web-based application developed to simplify and improve the day-to-day activities carried out in a pharmacy. In many pharmacies, maintaining medicine records, inventory details, customer information, and billing manually can be time-consuming and difficult to manage. This project provides a digital solution that helps organize pharmacy operations in a more efficient, accurate, and systematic manner. The application is developed using Python, Flask, HTML, CSS, and SQLite database technologies.

The main aim of the project is to reduce the manual effort involved in handling pharmacy records and to improve the overall efficiency of pharmacy management. The system includes important functionalities such as inventory management, medicine handling, customer management, and automated billing generation. By maintaining records digitally, the application helps reduce paperwork, minimize human errors, and improve the accuracy of daily pharmacy operations.

The Pharmacy Management System also helps pharmacists manage medicine stock effectively by updating inventory records automatically and tracking medicine availability. The billing module generates invoices quickly and accurately, which helps save time during customer transactions. Customer details and billing history can also be stored and retrieved easily whenever required.

SQLite database is used for secure and organized storage of pharmacy-related information, while Flask is used to establish smooth communication between the frontend interface and backend operations. The application is designed with a simple and user-friendly interface so that users can operate the system comfortably without requiring advanced technical knowledge.

Working on this project also provided practical knowledge about web application development, database management, frontend and backend integration, and real-world problem solving using software technologies. The project demonstrates how digital solutions can improve traditional pharmacy management methods and make operations more organized and reliable.

Overall, the Pharmacy Management System provides a secure, reliable, and efficient solution for managing pharmacy activities digitally while improving speed, accuracy, organization, and productivity within the pharmacy environment.

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CHAPTER 1

INTRODUCTION

1.1. PROBLEM DEFINITION

Managing a pharmacy manually can be a challenging and time-consuming task, especially when there are a large number of medicines, customers, and daily transactions to handle. In many pharmacies, records related to medicine stock, customer details, sales, and billing are maintained using paper registers or basic manual methods. As the pharmacy grows, handling all these records manually becomes difficult and increases the chances of mistakes. Errors such as incorrect billing, missing stock updates, duplicate entries, or loss of important information may occur frequently.

One of the major problems in manual pharmacy management is maintaining accurate inventory records. Pharmacists often find it difficult to keep track of medicine availability, expiry dates, and stock updates regularly. During busy hours, searching for medicine details or customer records from large amounts of paperwork can consume a lot of time and affect the quality of service provided to customers. Manual systems also make report generation and data retrieval more complicated and less efficient.

Another issue with traditional management methods is the lack of proper organization and security of records. Important data can easily be misplaced, damaged, or lost. Updating records manually on a daily basis also requires more effort and increases workload for pharmacy staff. In addition, maintaining sales information and customer purchase details manually can become confusing and difficult to manage over time.

To solve these problems, the Pharmacy Management System is developed as a digital and automated solution for managing pharmacy operations efficiently. The system helps in organizing all pharmacy-related activities in a simple and systematic manner. It allows users to add new medicines, update stock details, manage inventory, maintain customer records, and generate bills quickly and accurately. The system also helps in reducing paperwork and minimizing human errors by storing all records digitally.

The Pharmacy Management System improves the speed and accuracy of daily operations and makes data management easier for pharmacy staff. It provides faster access to information and helps in maintaining records securely and efficiently. The project is designed with a user-friendly interface so that users can operate the system easily without facing difficulties.

Overall, this project helps in improving pharmacy management by saving time, reducing manual effort, and increasing productivity. It provides a reliable and efficient solution for

handling pharmacy records and demonstrates how technology can be used to simplify real-world business operations.

1.2 OBJECTIVES

The main objective of the Pharmacy Management System is to simplify and automate the daily operations carried out in a pharmacy. The project is developed to reduce the difficulties involved in maintaining pharmacy records manually and to provide a more organized, reliable, and efficient method for handling pharmacy-related activities.

One of the important objectives of the system is to reduce paperwork and manual effort involved in maintaining medicine stock, billing information, supplier records, and customer details. By storing data digitally, the application helps in improving accuracy and reducing human errors that commonly occur in manual record management.

Another major objective of the project is to improve inventory management within the pharmacy. The system helps users maintain proper medicine stock details, update inventory records, monitor medicine availability, and track medicine expiry dates efficiently. It also makes it easier to manage medicine information such as quantity, price, manufacturer details, and supplier information.

The project also focuses on generating bills quickly and accurately. Automated billing reduces calculation errors and helps pharmacy staff complete transactions faster and more efficiently. Billing records are stored securely in the database, which helps in maintaining proper transaction history and financial records.

Customer record management is another important objective of the system. The application helps in storing, retrieving, and managing customer information easily, which improves organization, transaction tracking, and service management within the pharmacy.

The system is also designed to improve the overall workflow of pharmacy operations by integrating different modules such as inventory management, billing, customer management, sales handling, and supplier management into a single platform. This helps users access all important functionalities from one application conveniently.

Another objective of the project is to provide a simple, interactive, and user-friendly interface so that users can access different features without technical difficulty. Proper navigation and organized dashboard design improve accessibility and make the application easier to use for pharmacy staff.

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The project also aims to ensure secure storage and proper management of pharmacy data through database integration using SQLite. Faster data retrieval, improved data organization, and secure maintenance of records are additional goals of the system.

The Pharmacy Management System is further intended to save time, improve productivity, and reduce workload by automating repetitive pharmacy tasks. Digital record maintenance also improves reliability and reduces the risk of data loss compared to manual record handling.

Overall, the Pharmacy Management System aims to improve efficiency, accuracy, organization, and reliability while providing a complete digital solution for managing pharmacy operations effectively and systematically.

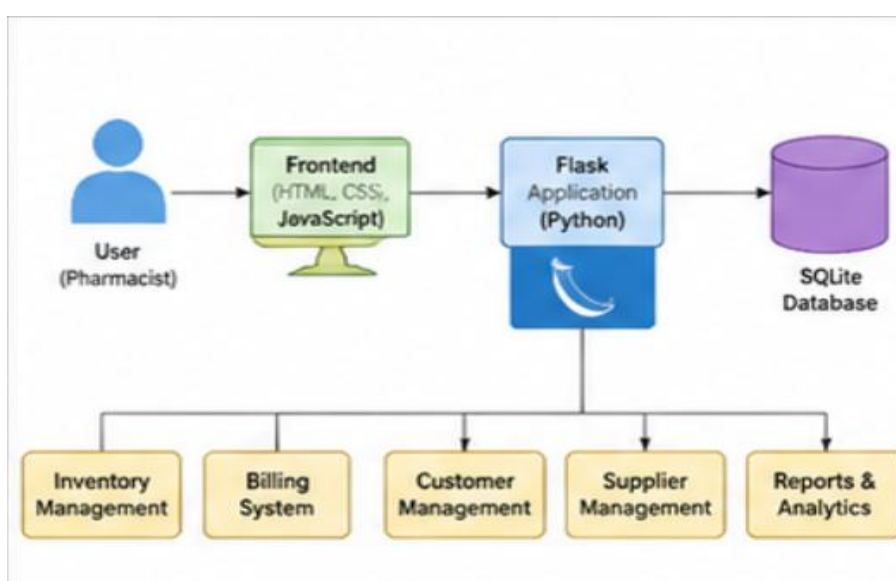


Figure 1.1 – Pharmacy Management System Architecture

1.3 METHODOLOGY TO BE FOLLOWED

The Pharmacy Management System is developed using Python, SQLite, HTML, CSS, and the Flask framework to create a simple, efficient, and user-friendly application for managing pharmacy activities digitally. Python is used for developing the main functionality of the system because it is easy to understand, flexible, and suitable for building management applications. SQLite is used as the database to store important information such as medicine details, inventory records, customer information, supplier details, sales transactions, and billing data securely.

HTML and CSS are used for designing the web pages and creating an interactive interface that is easy for users to access and understand. Flask helps in connecting the

frontend and backend of the application and allows smooth communication between the web pages and the database. It also helps in handling routing, form processing, and user requests efficiently.

The development of this project was carried out step by step in an organized and systematic manner. The first step was understanding the problems faced in manual pharmacy management, such as maintaining records, managing medicine stock, handling customer details, and generating bills manually. These issues often lead to calculation errors, data mismanagement, and increased paperwork.

After identifying the problems, the requirements of the system were analyzed carefully. Different functionalities required for the application, such as inventory management, customer management, billing system, supplier handling, and sales management, were planned based on the needs of pharmacy operations.

The next step involved designing the database structure and overall application workflow. Tables were created to store medicine details, customer records, billing information, supplier data, and inventory records. Proper relationships between different modules were designed to ensure smooth interaction within the system.

Once the design process was completed, different modules of the application were developed individually. The inventory module was created for managing medicine stock, the billing module was developed for invoice generation, and the customer management module was implemented for handling customer records efficiently.

The coding part of the project was implemented using Python, while database connectivity and query execution were handled using SQLite. Flask was used to manage backend operations and connect the frontend interface with database functionalities. SQL queries were used to insert, update, retrieve, and manage records within the database.

The user interface was designed in a simple, clean, and user-friendly manner so that users can operate the application easily without technical difficulty. Dashboard pages, navigation menus, forms, and tables were created to improve accessibility and make the application more interactive.

After completing the development process, the system was tested carefully to verify whether all modules and functionalities were working correctly. Different operations such as login authentication, medicine updates, customer record handling, billing generation, and inventory management were checked during testing. Errors and bugs identified during testing were corrected to improve the performance, reliability, and stability of the application.

Finally, the Pharmacy Management System was implemented successfully as a digital solution that helps in reducing manual work, saving time, improving accuracy,

maintaining organized records, and making pharmacy management more efficient, reliable, and systematic.

1.4 Expected Outcomes

The Pharmacy Management System is expected to improve the overall efficiency and organization of pharmacy operations by replacing traditional manual methods with a computerized solution. The system is designed to simplify daily pharmacy activities and provide faster, more accurate management of records, transactions, and inventory handling.

One of the major expected outcomes of the project is better inventory management. The system helps pharmacy staff maintain accurate medicine stock details, monitor medicine availability, track expiry dates, and update inventory records easily. This reduces the chances of stock shortages, duplicate entries, expired medicine usage, and incorrect inventory handling.

Another important outcome is faster and more accurate billing. The automated billing system calculates medicine prices, taxes, discounts, and total amounts efficiently, reducing manual calculation errors and improving transaction speed. This helps pharmacists serve customers more quickly and improves the overall billing experience.

The project is also expected to improve customer record management by storing customer details digitally in an organized manner. Customer information, billing history, and purchase records can be retrieved easily whenever required, which improves customer service and record maintenance within the pharmacy.

Efficient storage and management of pharmacy data is another major benefit of the system. All important information related to medicines, suppliers, inventory, billing, and customers is stored securely in the database, reducing paperwork and minimizing the risk of data loss or incorrect record handling.

The system is also expected to reduce manual effort and save time by automating repetitive pharmacy tasks. Faster data retrieval, improved accuracy, and proper organization of records help increase productivity and improve the workflow of pharmacy operations.

Another expected outcome is better financial and sales management. Since billing and sales records are maintained digitally, pharmacy staff can easily monitor transactions, sales performance, and purchase history whenever needed.

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The project is also expected to improve reliability and security of pharmacy data through proper database management and secure login authentication. This ensures that important records are protected from unauthorized access.

The Pharmacy Management System further aims to provide a user-friendly and interactive interface that can be used easily even by non-technical users. Proper navigation menus, dashboards, and organized modules improve usability and make pharmacy operations smoother and more systematic.

By integrating inventory management, billing, customer management, and database handling into a single application, the system is expected to provide a complete digital solution for efficient pharmacy management.

Overall, the Pharmacy Management System is expected to provide a reliable, organized, secure, and efficient solution for managing pharmacy activities digitally while improving accuracy, speed, productivity, and overall operational performance.

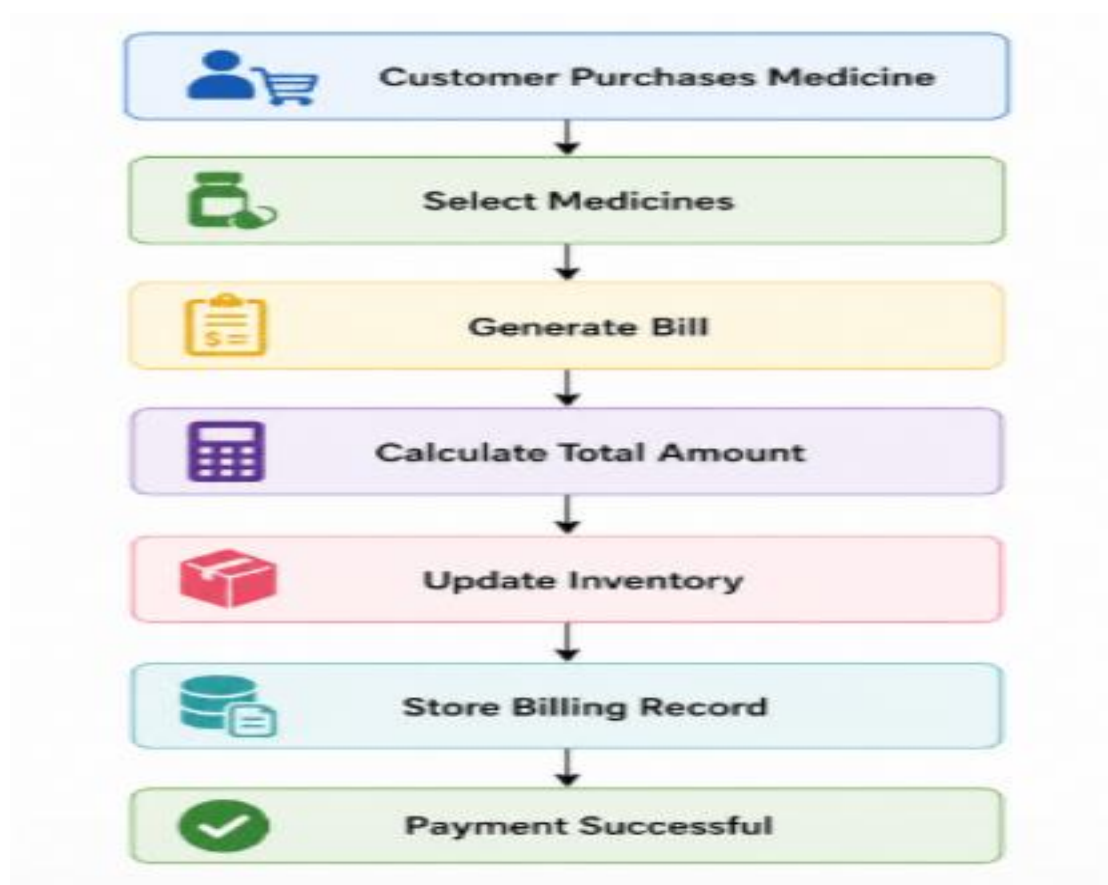


Figure 1.2 – Billing and Inventory Workflow

1.5 HARDWARE AND SOFTWARE REQUIREMENTS

The development of the Pharmacy Management System requires both hardware and software resources to ensure the smooth functioning of the application. A computer or laptop with a minimum of 4GB RAM and an Intel processor is used for developing and running the project efficiently. Adequate storage space is also necessary for storing the project files, database records, screenshots, reports, and other application-related data. A stable system configuration helps in carrying out coding, testing, debugging, and execution without interruptions or performance issues.

During the development process, different software tools and technologies are used to build the application successfully. Python is used as the main programming language because it is simple, easy to understand, and suitable for developing web-based applications. It also provides better flexibility, clean syntax, and faster development compared to many other programming languages.

SQLite is used as the database for storing important information such as medicine details, inventory records, billing information, supplier data, customer records, and sales transactions in an organized manner. Since SQLite is lightweight and easy to integrate with Python, it is suitable for mini projects and small-scale management systems like this application.

The frontend part of the project is designed using HTML and CSS templates to create a simple, responsive, and user-friendly interface. Flask framework is used to connect the frontend with the backend and to manage the flow of the application smoothly. Flask also helps in handling routing, form submissions, and communication between the user interface and the database.

Visual Studio Code (VS Code) is used as the development environment for writing, editing, debugging, and managing the project code efficiently. VS Code provides useful features such as syntax highlighting, extensions, terminal support, and code management, which make development easier and more organized.

A web browser such as Google Chrome is used to run and test the application interface during development. The browser helps in checking webpage responsiveness, functionality, and user interaction within the application.

The project can run on operating systems such as Windows, Linux, or macOS with minimal configuration changes. Internet connectivity may also be required for installing libraries, frameworks, and software packages needed during development.

In addition to these requirements, proper database connectivity and system compatibility are also important for the successful execution of the application. The hardware configuration should support smooth processing of billing operations, inventory updates, and data retrieval without delays. A reliable system environment also

helps developers test different modules efficiently and maintain application stability during execution.

The selected technologies are suitable for developing a pharmacy management application because they are lightweight, efficient, and easy to maintain. Python and Flask provide flexibility for backend development, while SQLite helps in secure and organized data storage. HTML and CSS improve the appearance and usability of the application by providing a clean and interactive user interface.

These hardware and software requirements together provide a proper platform for developing, testing, and executing the Pharmacy Management System effectively. The technologies used in the project help in improving the overall performance, reliability, usability, and efficiency of the application while making pharmacy management more organized, secure, and systematic.

Category	Hardware	Software
MINIMUM	Intel i3, 4GB RAM	Python, Flask, SQLite
Recommended	Intel i5, 8GB RAM	VS Code, Chrome Browser
OS	Windows/Linux	Windows/Linux/macOS

Table 1.1 – Hardware and Software Requirements

CHAPTER 2

FUNDAMENTALS OF PYTHON

2.1 Introduction to Python

Python is one of the most popular and widely used programming languages in the field of software and web application development. It is a high-level programming language known for its simple syntax, readability, and ease of understanding. Because of its beginner-friendly nature, Python is commonly used for developing applications, automation tools, data processing systems, and database-driven projects.

One of the major advantages of Python is that it allows developers to write programs with less code compared to many other programming languages. This makes the coding process simpler, faster, and more efficient. Python also provides a large number of built-in libraries and frameworks that help in developing applications easily and effectively.

In this project, Python is used to implement the backend functionality of the Pharmacy Management System. It helps in processing user requests, managing pharmacy operations, handling medicine and billing records, and connecting the application with the SQLite database. Python also supports database connectivity and file handling, which makes it suitable for developing management systems like this project.

The Pharmacy Management System uses Python to perform important operations such as adding new medicines, updating stock details, generating bills, storing customer information, and retrieving records from the database. Python helps in automating these operations and reduces the manual work involved in maintaining pharmacy data. It also improves the speed and accuracy of the overall system.

Another important feature of Python is its platform independence, which means the application can run on different operating systems with very few modifications. Python also supports object-oriented programming concepts that help in organizing the code in a structured and reusable manner. This makes the application easier to maintain, update, and expand in the future.

Python frameworks such as Flask make web application development easier by providing tools for routing, form handling, and database integration. In this project, Flask is used along with Python to connect the frontend interface with the backend logic and database operations. This helps in creating a smooth and user-friendly application for pharmacy management.

Python also provides better error handling and debugging features, which help developers identify and fix issues easily during application development. Its large community support and extensive documentation make learning and development more convenient for programmers.

Due to its flexibility, simplicity, and powerful features, Python is considered an ideal programming language for developing efficient and user-friendly applications. It plays an important role in making the Pharmacy Management System reliable, organized, secure, and easy to maintain. The use of Python in this project helps in improving efficiency, reducing errors, and simplifying pharmacy operations through digital automation.

2.2 Advantages of Python

Python is one of the most popular programming languages because it is simple, easy to understand, and convenient to use. Even beginners can learn Python quickly because its syntax is clear and similar to normal English statements. This makes coding easier and helps developers understand programs without much difficulty.

One of the main advantages of Python is that it is platform independent. A Python program can run on different operating systems such as Windows, Linux, and macOS with very few changes. This makes the language flexible and suitable for developing different kinds of applications.

Python also provides a large number of libraries and frameworks that make the development process faster and easier. These libraries contain ready-made functions that help developers perform tasks such as web development, database connectivity, automation, data processing, and file handling efficiently. In this project, Python is used along with Flask and SQLite to build the Pharmacy Management System smoothly and efficiently.

Another benefit of Python is that it requires fewer lines of code compared to many other programming languages. This reduces complexity and helps developers complete projects more quickly. Shorter and cleaner code also makes debugging and maintenance easier during development.

Python also supports object-oriented programming concepts, which help in organizing the code properly and make the application easier to maintain, modify, and reuse in the future. This is useful in projects like the Pharmacy Management System where multiple modules such as inventory management, billing, customer handling, and database operations are connected together.

Another important advantage of Python is its strong database connectivity support. Python can easily connect with databases like SQLite, which allows smooth storage,

retrieval, and management of pharmacy records such as medicine details, billing information, supplier records, and customer data.

Python also provides better error handling and debugging features, which help developers identify and fix issues more easily during application development. This improves the reliability and stability of the system.

The language is highly suitable for web application development because frameworks such as Flask simplify backend development and improve communication between the frontend and database. This helps in creating responsive and user-friendly applications.

Python has a huge community of developers and learners around the world. Because of this, there are many tutorials, guides, documentation resources, and support communities available for learning and problem-solving. Developers can easily find solutions and improve their projects using available resources.

Another advantage of Python is its scalability and flexibility. Applications developed using Python can be upgraded and expanded easily by adding new modules and functionalities in the future. This makes Python suitable for both small-scale and large-scale applications.

Due to its simplicity, flexibility, powerful features, and wide range of applications, Python is widely used for software development, web applications, automation, artificial intelligence, database systems, and many real-world projects. In the Pharmacy Management System, Python plays an important role in improving efficiency, reducing development complexity, and creating a reliable and organized digital solution for pharmacy management.

2.3 Data Types

Data types are used in Python to identify the type of data stored in a variable. They help the program understand whether the value is a number, text, or a collection of data. Different data types are used for different purposes depending on the kind of information that needs to be stored, processed, and managed within the application.

Python supports several basic data types such as integers, floats, strings, and booleans. Integer data types are used to store whole numbers like medicine quantity, customer IDs, stock count, or bill numbers. Float data types are used for decimal values such as medicine prices, discounts, GST percentages, and billing amounts where precise calculations are required.

Strings are used to store text information such as medicine names, customer names, supplier details, addresses, and other pharmacy-related records. They are one of the

most commonly used data types in management applications because most real-world information is stored in textual form.

Boolean data types store only two values, True or False, and are mainly used for checking conditions and making decisions within the program. For example, boolean values can be used to verify login authentication, check medicine availability, or confirm whether a transaction is completed successfully.

Python also provides collection data types like lists, tuples, sets, and dictionaries. Lists are used to store multiple values together and can be updated whenever needed. In the Pharmacy Management System, lists can be used to maintain medicine collections, sales records, or customer data temporarily during program execution.

Tuples are similar to lists but their values cannot be changed after creation. They are useful when fixed data needs to be stored securely without modification. Sets are used to store unique values without duplication, which helps in avoiding repeated entries.

Dictionaries store information in the form of key-value pairs, making data easier to organize and access efficiently. In this project, dictionaries can be useful for handling medicine details, billing data, and customer records where each value is connected with a specific key.

Using proper data types improves memory management, program accuracy, and overall application performance. Correct data handling also helps reduce errors during calculations, record management, and database operations.

In the Pharmacy Management System, different data types work together to manage medicine details, inventory records, billing information, supplier data, and customer transactions efficiently. Proper use of data types helps in organizing information systematically and makes the application more reliable, structured, and easier to maintain.

Overall, data types play an important role in application development by helping programmers store, process, and manage different kinds of information accurately and efficiently within the Pharmacy Management System.

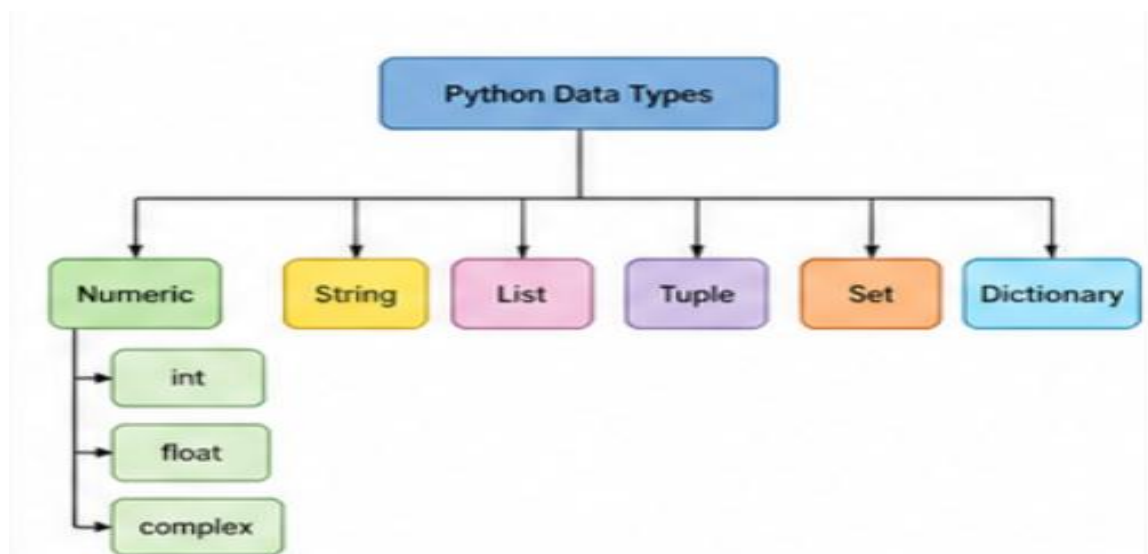


Figure 2.1 – Python Data Types Hierarchy

2.4 Python Numbers

Python provides different numeric data types that are used for performing mathematical and numerical operations in a program. These numeric data types help in storing and processing values such as quantities, prices, calculations, discounts, taxes, and other numerical information required in applications.

The most commonly used numeric data type in Python is the integer (int). Integers are used to store whole numbers without decimal values. In the Pharmacy Management System, integers can be used for storing medicine quantity, customer IDs, bill numbers, stock counts, and supplier IDs. Since these values do not require decimal points, integer data types are suitable for handling such records efficiently.

Another important numeric data type is the float (float). Float values are numbers that contain decimal points and are mainly used when precision is required. For example, medicine prices, discounts, GST calculations, billing totals, and sales amounts are usually stored using float data types. These values are important for generating accurate bills and maintaining proper financial records within the pharmacy.

Python also supports the complex (complex) data type, which is used to represent complex numbers containing real and imaginary parts. Although complex numbers are mostly used in scientific and engineering applications, Python provides built-in support for handling them easily. In the Pharmacy Management System, complex numbers are generally not required, but Python includes this feature as part of its numeric capabilities.

Python allows users to perform various mathematical operations such as addition, subtraction, multiplication, division, modulus, and exponentiation using these numeric data types. These operations are very useful in billing calculations, stock updates, profit calculations, and inventory management within the project.

The language also provides built-in mathematical functions and modules that simplify numerical calculations and improve program efficiency. Python can handle large calculations quickly and accurately, which makes it suitable for management systems that involve continuous data processing.

In the Pharmacy Management System, numeric data types play an important role in handling billing calculations, stock management, medicine pricing, inventory updates, sales records, and financial operations accurately and efficiently. Proper use of numeric data types helps reduce calculation errors and improves the reliability of the application.

Another advantage of Python numbers is that they are easy to use and require minimal coding effort. This helps developers create applications faster while maintaining better readability and simplicity in the program code.

Overall, Python numeric data types provide a reliable and efficient way of performing mathematical operations and managing numerical information within the Pharmacy Management System. They contribute significantly to maintaining accuracy, speed, and smooth functioning of pharmacy operations.

2.5 Python Strings

Strings in Python are used to store textual or character-based information. A string can contain letters, numbers, symbols, or a combination of all of them, and it is usually written within single quotes or double quotes. Strings are one of the most commonly used data types in Python because many applications work mainly with text-related information.

In the Pharmacy Management System, strings are used for storing details such as medicine names, customer names, supplier details, addresses, contact numbers, email IDs, bill information, and other text-based records. They help in managing and displaying information clearly and systematically within the application.

Python provides many built-in functions and operations for working with strings. Users can join strings together, search for specific words, replace text, convert characters into uppercase or lowercase, remove unnecessary spaces, and find the length of a string easily. These string operations help developers process and manage textual data more efficiently.

Another important feature of Python strings is indexing and slicing. Indexing allows access to individual characters in a string, while slicing helps in extracting a part of the string whenever required. These features are useful for handling and processing text data accurately within applications.

Strings also support formatting operations, which help developers display information in a more readable and organized way. In the Pharmacy Management System, formatted strings can be used while generating bills, displaying medicine details, showing customer information, and presenting transaction records properly on the user interface.

Strings play an important role in creating user-friendly applications because most of the information entered by users is in text format. Medicine descriptions, supplier information, login credentials, search operations, and billing details are all managed using strings within the system.

Another advantage of Python strings is that they are easy to handle and highly flexible. Python provides many built-in methods that simplify text processing and reduce coding complexity. This helps developers create applications faster and maintain the code more easily.

In the Pharmacy Management System, strings are widely used in forms, dashboard displays, billing pages, customer records, inventory management, and database operations. Proper handling of string data helps improve accuracy, readability, and organization within the application.

Overall, Python strings provide an efficient and reliable way of storing and managing textual information. They play an important role in handling pharmacy-related data and contribute to making the Pharmacy Management System more organized, interactive, and user-friendly.

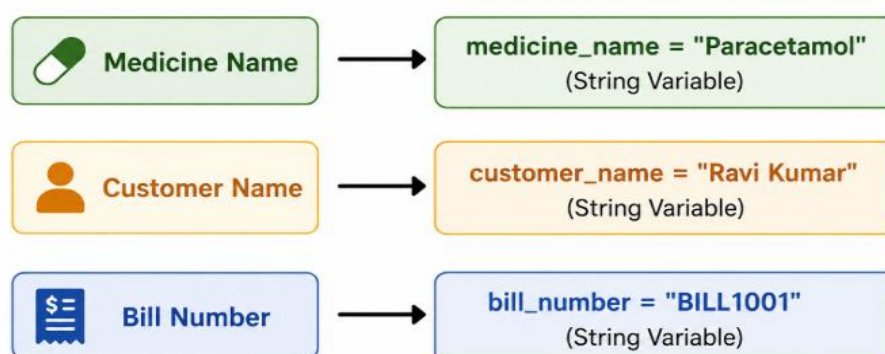


Figure 2.2 – String Handling in Pharmacy System

2.6 Python Lists

Lists in Python are used to store multiple values together in a single variable. They are one of the most commonly used collection data types because they allow data to be stored in an organized and flexible manner. Lists are created using square brackets and can contain different types of data such as numbers, strings, or even other lists.

One of the main features of Python lists is that they are mutable, which means their values can be changed, added, or removed whenever required. This makes lists very useful in applications where data needs to be updated frequently.

In the Pharmacy Management System, lists can be used to manage medicine details, inventory records, customer information, and billing data. For example, a list can store multiple medicine names, stock quantities, or customer details together for easy access and processing.

Python provides many built-in functions and methods for working with lists. Elements can be added using `append()`, removed using `remove()`, and updated whenever necessary. Lists also support indexing and slicing, which makes it easy to access specific values from the collection.

Another advantage of lists is that they help in reducing the complexity of handling large amounts of data. Instead of creating separate variables for each value, multiple values can be stored and managed efficiently within a single list.

Due to their flexibility and ease of use, lists play an important role in Python programming and are widely used in developing real-world applications like the Pharmacy Management System.

2.7 Python Tuples

Tuples in Python are used to store multiple values together in a single variable, similar to lists. However, unlike lists, tuples are immutable, which means their values cannot be modified once they are created. Tuples are written using round brackets and can contain different types of data such as numbers, strings, and other values.

Tuples are mainly used when the data should remain fixed and should not be changed during program execution. Because of this feature, tuples provide better data security and help in preventing accidental modifications to important information.

In the Pharmacy Management System, tuples can be used for storing fixed records or predefined data that does not need frequent updates. For example, tuples may be used

to store medicine categories, tax percentages, fixed configuration values, supplier codes, or predefined menu options within the application.

Python allows users to access tuple elements using indexing, similar to lists. Although values inside a tuple cannot be changed, the data can still be read and processed easily whenever required. This makes tuples useful for handling secure and constant information within the program.

Another advantage of tuples is that they consume less memory compared to lists, making them more efficient in certain situations. Since tuples are immutable, Python can process them faster and more securely than some other collection types.

Tuples also improve program reliability because the stored data remains unchanged throughout execution. This helps maintain consistency in applications where fixed data is important and should not be modified accidentally during operations.

In the Pharmacy Management System, tuples can help maintain stable records for predefined medicine categories, supplier details, billing constants, or fixed application settings. Using tuples for such data improves system consistency and reduces the chances of unwanted changes.

Tuples are also useful when multiple values need to be returned from a function together. In Python, functions can return tuples containing several related values at once, which helps simplify program logic and improve code organization.

Another important benefit of tuples is improved readability and cleaner code structure. Since tuples are used mainly for fixed data, developers can easily identify information that should remain constant throughout the application.

Due to their simplicity, efficiency, and reliability, tuples are widely used in Python programming for storing secure and unchangeable collections of data. They help developers create more stable and organized applications.

Overall, tuples play an important role in Python programming by providing a simple and efficient way of storing fixed information. In the Pharmacy Management System, tuples contribute to better data consistency, improved reliability, and organized management of predefined records within the application.

2.8 Python Sets

Sets in Python are used to store collections of unique values. Unlike lists and tuples, sets do not allow duplicate elements, which makes them useful when only distinct values

need to be stored. Sets are written using curly brackets and can contain different types of data such as numbers, strings, or other immutable values.

One important feature of sets is that they are unordered, which means the elements are not stored in a fixed sequence. Because of this, sets are mainly used for operations where uniqueness of data is more important than maintaining order. This makes sets useful for filtering repeated information and managing non-duplicate records efficiently.

In the Pharmacy Management System, sets can be useful for storing unique medicine categories, supplier names, customer IDs, medicine brands, or other records where duplicate values should be avoided. Using sets helps maintain clean, organized, and non-repetitive data within the application.

Python provides different operations for working with sets, such as adding elements, removing elements, and performing mathematical set operations like union, intersection, and difference. These operations make sets useful for comparing and filtering data efficiently during application processing.

For example, set operations can help identify common medicines between different suppliers or compare available stock categories. They can also be useful while handling unique medicine records or checking repeated entries in inventory management.

Another advantage of sets is that they provide faster searching and checking of values compared to some other collection types. Since duplicate entries are automatically removed, sets help reduce data redundancy and improve data management within the application.

Sets are also easy to modify because elements can be added or removed whenever required. This flexibility makes them suitable for handling dynamic data where records may change frequently during program execution.

In the Pharmacy Management System, sets help improve efficiency while handling unique pharmacy-related information. They support better organization of records and help avoid unnecessary duplication in the database or temporary data handling operations.

Although sets are not used for ordered storage, they are highly useful in situations where uniqueness, faster processing, and efficient searching are important. Their simple syntax and built-in operations make them easy to use within Python programs.

Overall, Python sets provide a simple and effective way of managing unique and non-repetitive data. They contribute to better data handling, improved efficiency, and more organized processing within the Pharmacy Management System.

2.9 Python Dictionaries

Dictionaries in Python are used to store data in the form of key-value pairs. They are one of the most useful and flexible data structures in Python because they allow data to be organized, managed, and accessed efficiently. Dictionaries are created using curly brackets, where each key is associated with a specific value.

In a dictionary, keys are unique and are used to identify the corresponding values easily. This makes dictionaries very useful when handling structured information and managing related data together. Since each value is connected with a unique key, retrieving information becomes faster and more organized.

For example, in the Pharmacy Management System, dictionaries can be used to store medicine details such as medicine name, price, quantity, manufacturer details, and expiry date in a systematic manner. They can also help manage customer records, billing information, and supplier details efficiently.

One of the major advantages of dictionaries is that values can be accessed quickly using their keys instead of searching through the entire collection. This improves processing speed and makes data handling more efficient, especially in applications that work with large amounts of information.

Dictionaries also allow data to be added, updated, or removed whenever required, making them highly flexible for real-world applications. In the Pharmacy Management System, medicine stock details or customer information can be modified easily whenever changes occur in the database or application records.

Python provides several built-in methods for working with dictionaries, such as adding new entries, updating existing values, retrieving data, checking available keys, and deleting unwanted records. These operations help in managing large amounts of pharmacy-related data more efficiently and accurately.

Another important advantage of dictionaries is that they help improve code readability and organization. Since information is stored in a structured key-value format, developers can understand and manage data more easily during program development and maintenance.

Dictionaries are especially useful when related information needs to be grouped together logically. For example, a single dictionary can store complete details about a medicine or customer, making record management simpler and more organized.

In this project, dictionaries help in organizing pharmacy-related information clearly and systematically. They improve data management and make it easier to handle customer records, medicine details, inventory information, billing data, and supplier records within the application.

The flexibility and efficiency of dictionaries make them highly suitable for database-driven applications like the Pharmacy Management System, where fast access and proper organization of records are important.

Overall, Python dictionaries provide a powerful and efficient way of storing and managing structured data. They contribute significantly to improving data organization, processing speed, and overall functionality within the Pharmacy Management System.

2.10 Functions in Python

Functions in Python are used to group a set of statements together to perform a specific task. They help in organizing the program into smaller and manageable parts, making the code easier to understand, maintain, and debug. Instead of writing the same code repeatedly, functions allow developers to reuse code whenever needed, which improves programming efficiency.

A function is created using the `def` keyword followed by the function name and parentheses. Once a function is defined, it can be called multiple times from different parts of the program. This reduces code repetition and improves the overall structure and readability of the application.

In the Pharmacy Management System, functions are used to perform various operations such as adding medicine details, updating stock information, generating bills, managing customer records, handling supplier data, and retrieving information from the database. By dividing the program into functions, the application becomes more organized and easier to manage.

Functions help developers separate different modules and functionalities clearly. For example, separate functions can be created for login authentication, inventory updates, billing calculations, customer management, and database connectivity. This modular approach improves code organization and simplifies application development.

Another important advantage of functions is code reusability. A single function can be reused multiple times whenever the same operation is required. For example, a billing calculation function can be called every time a new invoice is generated, which saves development time and reduces unnecessary coding.

Functions also make debugging and testing easier because errors can be identified and corrected within a specific part of the program without affecting the entire application. This improves the reliability and stability of the system.

Python functions can also accept parameters and return values. Parameters allow functions to process different inputs, while return statements help functions send

results back to the program. In the Pharmacy Management System, functions may return billing totals, medicine details, inventory updates, or customer records based on user requests.

Using functions improves the overall maintainability of the application because modifications can be made to individual functions without rewriting the complete program. This is especially useful for projects that may require future updates and additional features.

Functions also improve program readability because the code becomes more structured and easier to understand. Developers can quickly identify the purpose of each function based on its name and functionality.

In the Pharmacy Management System, functions play an important role in simplifying backend operations and ensuring smooth execution of different pharmacy-related activities. They help in managing application flow efficiently and support better interaction between the frontend, backend, and database.

Overall, functions are an essential part of Python programming because they improve code organization, reusability, efficiency, and maintainability. They contribute significantly to building a reliable, structured, and efficient Pharmacy Management System.

Functions also improve code readability because each function performs a particular task independently. If any changes or corrections are required, they can be made easily within the specific function without affecting the entire program. This makes debugging and maintenance simpler and more efficient.

Python supports different types of functions such as built-in functions, user-defined functions, and functions with parameters and return values. These features provide flexibility while developing applications.

Overall, functions play an important role in improving code modularity, re-usability, and efficiency. They help in developing structured and reliable applications like the Pharmacy Management System in a simpler and more effective way.

CHAPTER 3

FUNDAMENTALS OF DBMS

3.1 Introduction

A Database Management System (DBMS) is software used to store, organize, manage, and retrieve data efficiently. It helps users handle large amounts of information in a systematic, secure, and reliable manner. Instead of maintaining records manually using files, notebooks, or registers, a DBMS allows data to be stored digitally, making data management easier, faster, and more organized.

In many organizations and applications, huge amounts of data are generated every day. Managing this data manually can be difficult, time-consuming, and prone to errors. A DBMS helps overcome these problems by providing proper methods for storing, updating, searching, and accessing data whenever required. It also reduces data redundancy and improves data consistency by organizing records properly within the database.

In the Pharmacy Management System, the DBMS plays an important role in storing medicine details, customer information, inventory records, supplier data, sales history, and billing information. The database helps organize all pharmacy-related information properly so that it can be accessed quickly and efficiently whenever needed.

Another important advantage of a DBMS is data security and reliability. It helps protect important information from unauthorized access and prevents accidental loss or modification of records. Secure database management is important in applications like pharmacy systems where sensitive customer and billing data are maintained digitally.

A DBMS also allows multiple operations such as adding, updating, deleting, searching, and retrieving records in a simple and efficient manner. These operations help pharmacy staff manage daily activities more effectively without depending on manual record maintenance.

The database system also improves the speed and accuracy of pharmacy operations. Users can quickly search for medicine details, check stock availability, retrieve billing records, and manage customer information without spending extra time searching through physical files.

SQLite is used as the Database Management System in this project because it is lightweight, easy to use, and suitable for small and medium-scale applications. SQLite

integrates easily with Python and Flask, which makes database connectivity and data handling simpler during application development.

Another advantage of using a DBMS in the Pharmacy Management System is better organization of records. Medicine stock details, billing data, and customer information are maintained systematically, which improves workflow management and overall efficiency within the pharmacy.

The DBMS also supports future scalability of the application. Additional features and records can be added easily without changing the entire system structure. This flexibility makes the system easier to maintain and upgrade in the future.

Overall, a Database Management System provides an effective and reliable solution for managing large amounts of data in an organized manner. It plays a major role in developing efficient, secure, and user-friendly applications like the Pharmacy Management System by improving accuracy, reducing manual work, and simplifying data management operations.

3.2 Characteristics of a DBMS

A Database Management System (DBMS) provides several important features that help in storing and managing data efficiently. It is designed to organize data in a structured manner and make data handling easier, faster, more secure, and more reliable. These characteristics make a DBMS an essential part of modern software applications and management systems.

One of the main characteristics of a DBMS is data consistency. It ensures that the data stored in the database remains accurate and uniform throughout the system. Whenever changes are made to the data, the updates are reflected properly without creating confusion or incorrect information. This helps maintain reliable records within the application and improves the accuracy of operations.

Another important feature of a DBMS is reduced data redundancy. In manual systems, the same data may be stored repeatedly in multiple places, leading to duplication and wastage of storage space. A DBMS minimizes duplicate data by organizing information systematically and storing it efficiently. This helps improve database organization and reduces unnecessary repetition of records.

Data security is also an important characteristic of a DBMS. It helps protect sensitive information from unauthorized access and ensures that only authorized users can view or modify the data. In the Pharmacy Management System, security is important for

protecting medicine records, billing information, supplier details, and customer data from misuse or accidental modification.

A DBMS also provides efficient data retrieval. Users can quickly search, access, and retrieve information whenever required without manually searching through files or registers. This improves the speed of pharmacy operations and helps users access records more conveniently.

Another important characteristic is data sharing and multi-user support. A DBMS allows multiple users to access and manage the database simultaneously without affecting data accuracy or consistency. This feature is useful in environments where multiple pharmacy staff members may work on billing, inventory management, or customer handling at the same time.

Backup and recovery are also important features of a DBMS. The system helps maintain data safety by allowing recovery of records in case of accidental deletion, software issues, or system failures. This improves the reliability and security of important pharmacy records.

Data integrity is another key characteristic of a DBMS. It ensures that only valid and correct data is stored in the database. Proper validation rules and constraints help prevent incorrect entries and maintain high-quality records within the system.

A DBMS also improves data organization by storing records in a structured format using tables, rows, and columns. This organized structure makes the database easier to manage and simplifies record maintenance.

In the Pharmacy Management System, these DBMS characteristics help manage medicine stock, customer details, supplier information, billing records, and sales transactions more effectively. They improve efficiency, reduce manual effort, and support smooth functioning of the application.

Overall, the characteristics of a DBMS help create reliable, secure, organized, and efficient systems for handling large amounts of data. These features play a major role in improving the performance and usability of the Pharmacy Management System.

A DBMS also supports easy data retrieval and management. Users can quickly search, update, insert, or delete records whenever required. This improves the speed and efficiency of the application and reduces manual effort.

Another advantage of DBMS is data backup and recovery. In case of system failure or accidental data loss, the database can help recover important information. This increases the reliability and safety of the stored records.

In the Pharmacy Management System, these characteristics help in maintaining pharmacy records accurately, managing inventory efficiently, and improving the overall performance of the application.

3.3 Data Model

A data model is a method used to organize and structure data in a database. It defines how data is stored, connected, and managed within the system. Data models help represent real-world objects and their relationships in a simple and understandable manner. They also provide a clear structure for designing, maintaining, and managing databases efficiently.

Different types of data models are used in database systems, such as hierarchical, network, relational, and object-oriented data models. Among these, the relational data model is one of the most commonly used models because of its simplicity, flexibility, and efficiency in handling structured data.

In this project, the relational data model is used for designing the Pharmacy Management System database. In the relational model, data is stored in the form of tables consisting of rows and columns. Each table represents a specific entity such as medicines, customers, suppliers, inventory, billing details, or sales records.

The rows in a table represent individual records, while the columns represent attributes or fields related to those records. For example, a medicine table may contain columns such as medicine ID, medicine name, category, quantity, manufacturer details, price, and expiry date. This tabular structure helps organize data clearly and systematically.

Relationships can also be established between different tables using primary keys and foreign keys. This allows related information to be connected and accessed easily whenever required. For instance, billing records can be connected with customer details and medicine information through relational keys.

One of the major advantages of the relational data model is reduced data redundancy. Since data is organized into separate related tables, duplicate information is minimized and storage becomes more efficient. This also improves data consistency and accuracy throughout the system.

The relational model also supports efficient data retrieval using SQL queries. Operations such as inserting, updating, deleting, and searching records can be performed easily and quickly. This improves the speed and reliability of database operations within the Pharmacy Management System.

Another important benefit of the relational model is better data organization and maintainability. Since records are stored systematically in tables, developers and users can understand and manage the database structure more easily.

In the Pharmacy Management System, the relational data model helps manage medicine stock, customer information, supplier details, billing records, inventory updates, and sales transactions efficiently. It ensures that all pharmacy-related information is properly connected and maintained within the database.

The relational data model also improves scalability of the application because new tables, records, or features can be added easily in the future without affecting the overall database structure significantly.

Overall, the relational data model provides a simple, organized, and reliable way of storing and managing data. It plays an important role in improving database efficiency, reducing complexity, and supporting smooth operation of the Pharmacy Management System.

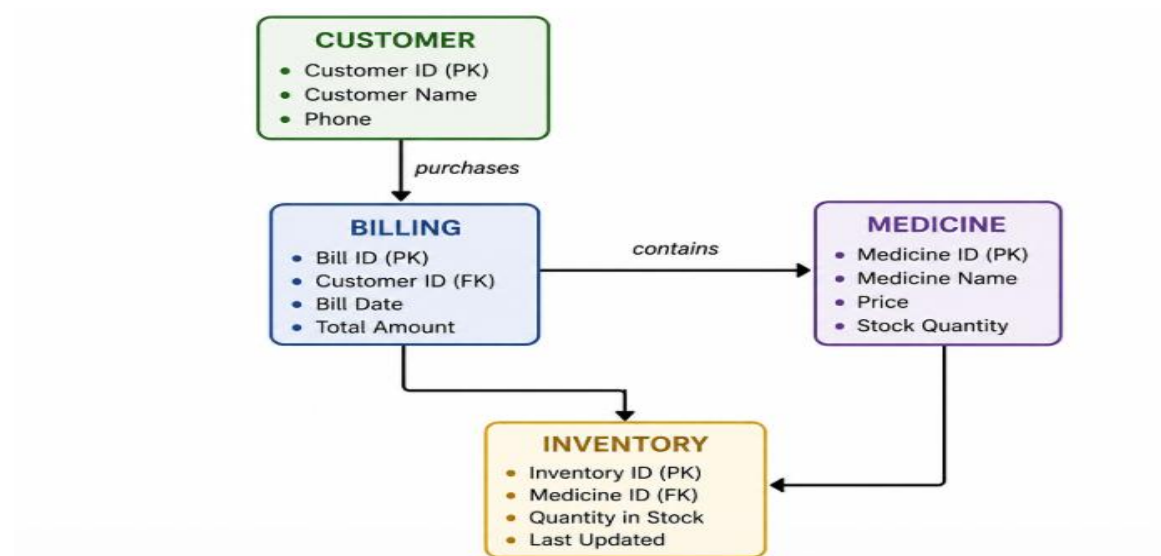


Figure 3.1 – ER Diagram of Pharmacy Management System

3.4 Three Schema Architecture

The Three Schema Architecture is a database architecture used to separate the database system into different levels of abstraction. It helps in organizing the database structure properly and makes data management easier, more secure, and flexible. This architecture improves data independence and allows users to interact with the database without needing to know the complete internal details of how the data is stored.

The Three Schema Architecture consists of three different levels: External Level, Conceptual Level, and Internal Level. Each level performs a specific role in the database system and helps in managing data efficiently.

The External Level is the topmost level of the architecture and represents how users view the data. Different users may require different views of the same database depending on their needs and responsibilities. For example, in the Pharmacy Management System, a pharmacist may access medicine stock and inventory details, while another user may work mainly with billing information and customer records. This level displays only the required information to the user and hides unnecessary database details, making the system simpler and easier to use.

The Conceptual Level is the middle level of the architecture and describes the complete logical structure of the database. It contains information about all the entities, relationships, tables, attributes, and constraints used in the system. In the Pharmacy Management System, this level includes details related to medicines, inventory, billing, customer records, and sales transactions. The conceptual level acts as a bridge between the external and internal levels and provides an overall view of the entire database structure.

The Internal Level is the lowest level of the architecture and deals with the physical storage of data inside the system. It defines how the data is actually stored in memory, including file structures, storage allocation methods, indexing, and data retrieval techniques. Users generally do not interact directly with this level because it mainly focuses on efficient storage and performance management.

In the Pharmacy Management System, the Three Schema Architecture helps in organizing pharmacy data in a systematic and structured manner. It allows smooth interaction between users and the database while maintaining proper security and data consistency. The architecture also helps in reducing complexity because changes made at one level do not directly affect the other levels.

Another important advantage of this architecture is data independence. For example, changes in the physical storage of the database do not affect how users view the information through the application. This makes the system easier to maintain, modify, and upgrade in the future without causing major issues.

The Three Schema Architecture also improves scalability and reliability of the database system. As the pharmacy grows and more records are added, the database can still manage data efficiently without affecting system performance. Overall, this architecture plays an important role in making the Pharmacy Management System organized, secure, flexible, and efficient for handling pharmacy operations digitally.

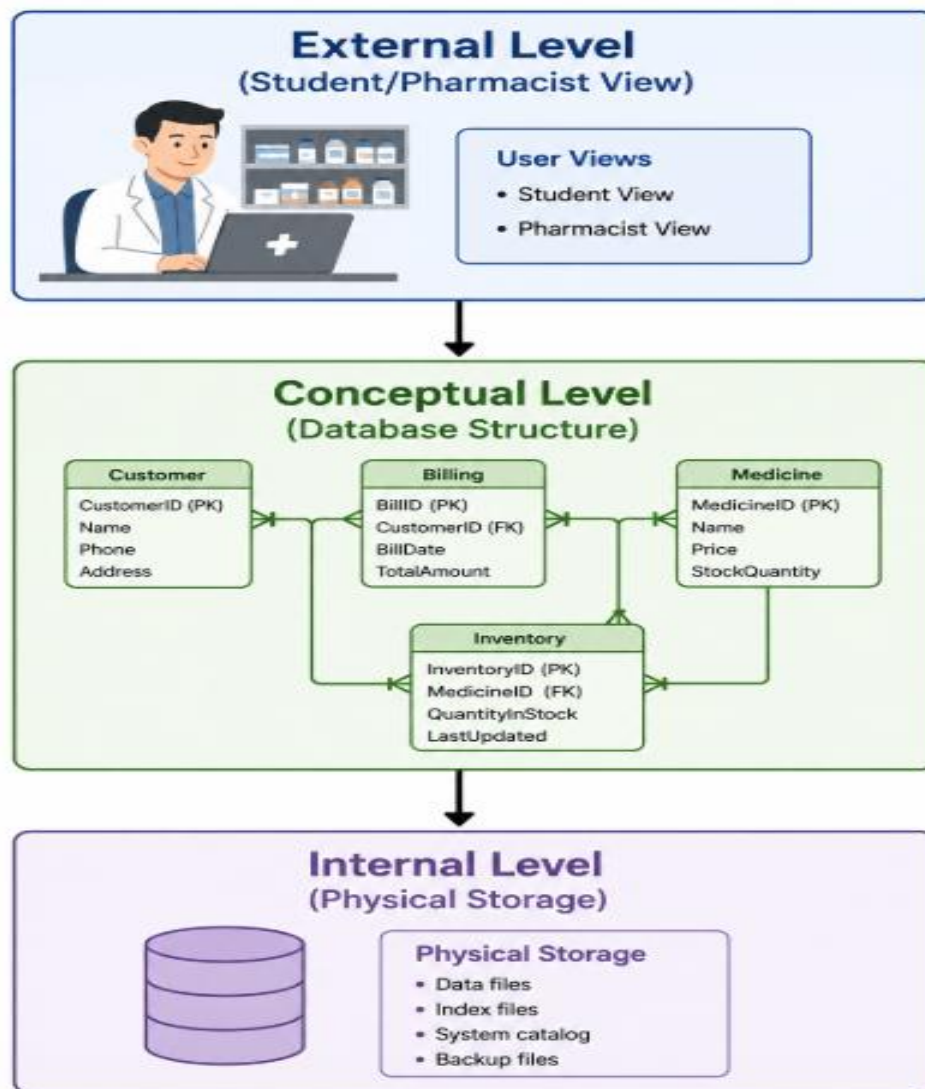


Figure 3.2 – Three Schema Architecture

3.5 DBMS Components

A Database Management System (DBMS) consists of several important components that work together to store, manage, and retrieve data efficiently. These components help in processing user requests, maintaining data security, and ensuring smooth functioning of the database system. Each component performs a specific task that contributes to the overall performance and reliability of the application.

One of the main components of a DBMS is the Query Processor. The query processor is responsible for handling database queries entered by the user. It interprets commands such as inserting, updating, deleting, or retrieving data from the database. The query

processor converts these commands into a form that the database system can understand and execute efficiently. In the Pharmacy Management System, the query processor helps execute operations related to medicine records, billing details, customer management, and inventory updates.

Another important component is the Storage Manager. The storage manager controls how data is stored in the database and manages memory allocation, file organization, and data retrieval. It ensures that records are stored properly and can be accessed quickly whenever required. In the Pharmacy Management System, the storage manager helps organize medicine details, supplier records, customer information, and billing data efficiently within the database.

The Transaction Manager is another essential component of a DBMS. It manages database transactions and ensures that all operations are completed accurately and consistently. A transaction may include multiple database operations such as updating stock information and generating bills together. The transaction manager helps maintain data integrity and prevents errors during system failures or unexpected interruptions.

Another important component is the Data Dictionary, which stores information about the database structure, tables, columns, relationships, and constraints. It acts as a reference guide for the database and helps manage metadata efficiently.

The Authorization Manager is responsible for maintaining database security and controlling user access. It ensures that only authorized users can access or modify sensitive information. In the Pharmacy Management System, this component helps protect medicine records, customer details, and billing information from unauthorized access.

The Backup and Recovery Manager helps maintain data safety by creating backups and recovering records in case of accidental deletion, hardware failure, or software issues. This improves the reliability and security of the database system.

The File Manager is another component that manages database files and controls how records are stored physically in memory. It helps improve data organization and efficient file handling within the system.

All these DBMS components work together to provide smooth and efficient database operations. They help manage large amounts of information systematically and ensure better performance, security, and reliability.

In the Pharmacy Management System, DBMS components play an important role in storing pharmacy records, processing user requests, managing transactions, handling billing operations, and maintaining organized inventory data. These components help improve system efficiency and simplify overall pharmacy management.

Overall, the components of a DBMS contribute significantly to building a secure, reliable, and efficient database system. They help maintain smooth functioning of the Pharmacy Management System while ensuring proper data handling, faster retrieval, and accurate record management.

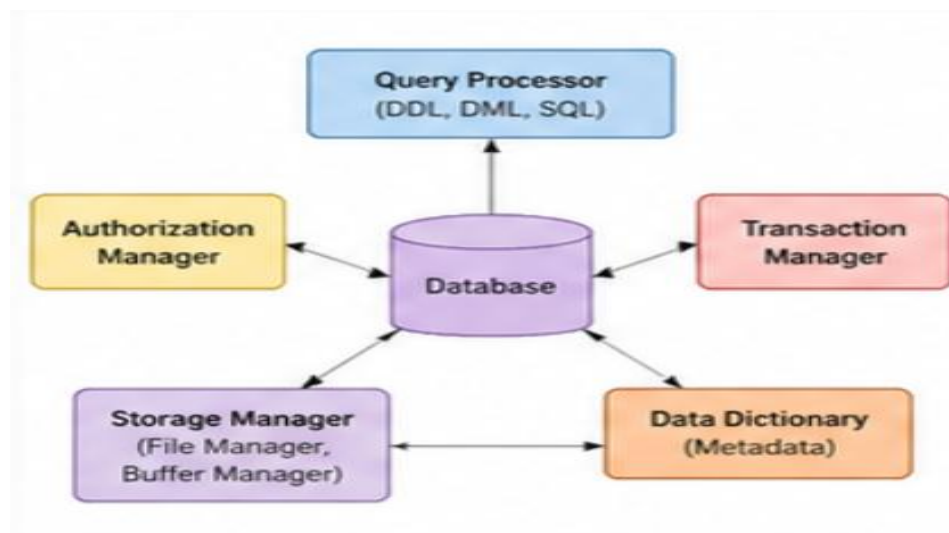


Figure 3.3 – DBMS Components

3.6 Entity Relationship Model

The Entity Relationship (ER) Model is a conceptual model used to design and represent the structure of a database. It helps in identifying the different entities involved in the system and the relationships between them. The ER model provides a clear and simple way of understanding how data is organized within a database before the actual database implementation process begins.

In an ER model, an entity represents a real-world object or item about which data is stored. Each entity contains attributes that describe its properties and characteristics. For example, in the Pharmacy Management System, entities such as Medicines, Customers, Inventory, Billing, and Suppliers are used to represent the major components of the application.

The Medicine entity contains important details related to medicines available in the pharmacy. It may include attributes such as medicine ID, medicine name, quantity, price, manufacturer details, category, and expiry date. These details help in maintaining accurate stock information and tracking medicine availability within the pharmacy.

The Customer entity stores information related to customers who purchase medicines from the pharmacy. This entity may contain attributes such as customer ID, customer

name, contact number, email address, and residential address. Managing customer records digitally helps in maintaining purchase history and improving customer service.

The Billing entity stores information related to sales transactions and invoice generation. It may include bill number, billing date, purchased medicine details, total amount, discount information, and payment method. The billing module helps in generating bills accurately and maintaining proper transaction records.

The Inventory entity is used to manage medicine stock and monitor availability. It helps in tracking medicine quantity, stock updates, reorder levels, and recently updated records. Proper inventory management reduces the chances of stock shortages and improves the efficiency of pharmacy operations.

Relationships are established between entities to show how they are connected with each other. For example, a customer can purchase multiple medicines, and billing records are connected with both customer and medicine details. Similarly, inventory records are directly related to medicine information. These relationships help in organizing data systematically and maintaining proper connections between different records in the database.

The ER model also helps in reducing data redundancy and improving data consistency by clearly defining the structure of the database before implementation. Since the relationships and attributes are planned properly in advance, the chances of duplicate or inconsistent data are minimized.

Another important advantage of the ER model is that it makes database design easier to understand for both developers and users. It provides a visual representation of the database structure, which helps in identifying how different modules of the system interact with each other.

In the Pharmacy Management System, the ER model plays an important role in planning and organizing the database structure efficiently. It acts as a blueprint for creating database tables, defining relationships, and managing pharmacy records systematically. Overall, the ER model helps in developing a well-structured, organized, and efficient database system that supports smooth pharmacy management operations.

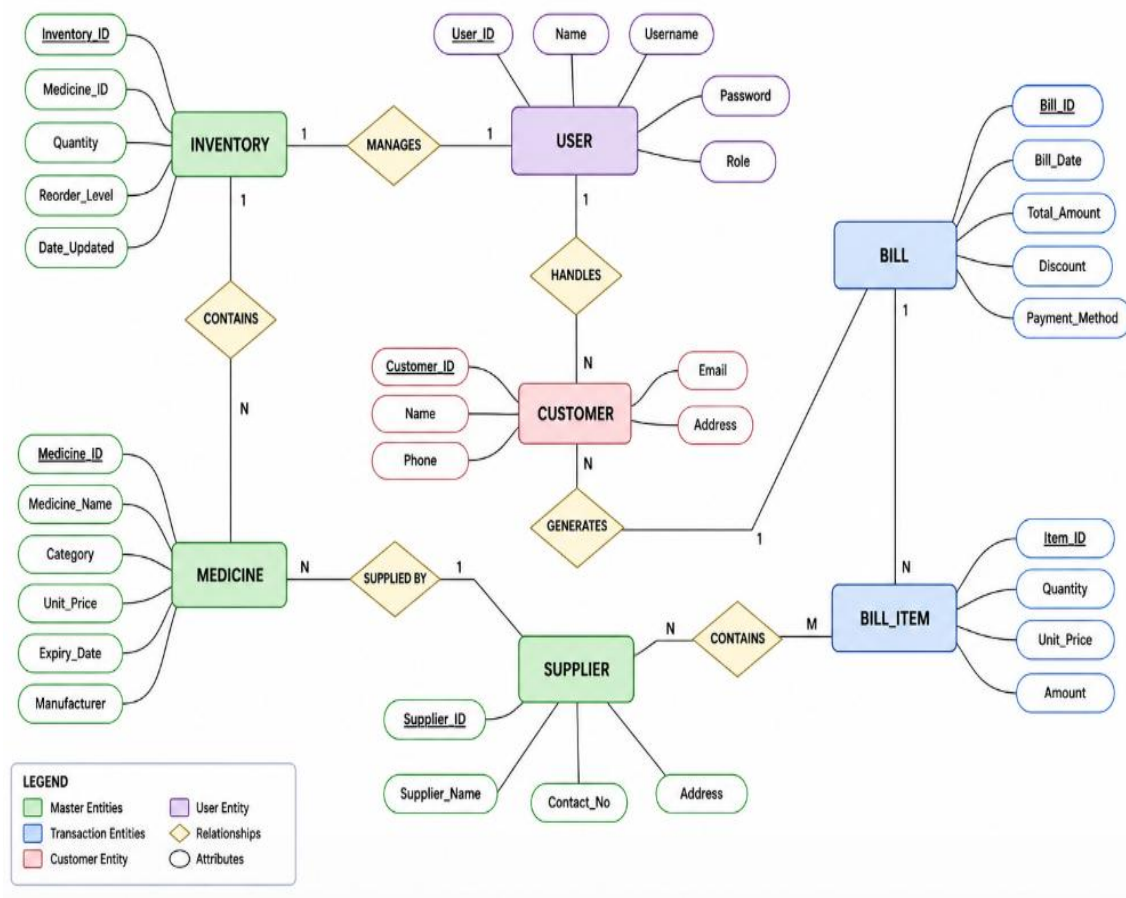


Figure 3.4 – ER Diagram

3.7 Relational Schema

A relational schema is a structure that defines how data is organized in a relational database. It describes the tables, columns, data types, and relationships used in the database system. The relational schema acts as a blueprint for the database and helps organize data in a clear, systematic, and well-structured manner.

In a relational database, data is stored in the form of tables. Each table represents a specific entity, and the information is arranged in rows and columns. Rows represent individual records, while columns represent the attributes or fields related to those records. This tabular structure makes the database easy to understand, maintain, and manage efficiently.

In the Pharmacy Management System, different tables are created to store various types of information such as medicine details, customer records, supplier information,

inventory data, sales records, and billing details. Each table is designed carefully to store related information systematically within the database.

For example, the medicine table may contain fields such as medicine ID, medicine name, category, quantity, price, manufacturer details, supplier information, and expiry date. Similarly, the customer table stores customer-related information such as customer ID, customer name, phone number, email address, and purchase history.

The billing table may include bill number, customer details, medicine information, quantity purchased, GST amount, total price, and payment details. Inventory tables help maintain stock availability and medicine updates efficiently within the pharmacy system.

Relationships are established between tables using keys such as primary keys and foreign keys. Primary keys uniquely identify each record in a table, while foreign keys help connect related tables together. These relationships improve data consistency and allow related information to be accessed efficiently whenever required.

One of the major advantages of a relational schema is reduced data redundancy. Since related information is stored in separate connected tables, duplicate data is minimized and database organization becomes more efficient.

The relational schema also helps simplify database operations such as insertion, deletion, updating, and retrieval of records. SQL queries can be executed easily to access and manage information stored within the database tables.

Another important benefit of the relational schema is improved data integrity and consistency. Since relationships and constraints are defined clearly, the chances of storing incorrect or duplicate information are reduced significantly.

In the Pharmacy Management System, the relational schema plays an important role in maintaining pharmacy records properly and ensuring smooth database operations throughout the application. It helps manage medicine stock, customer details, billing information, supplier records, and inventory data efficiently.

The relational schema also improves scalability and maintainability of the system. Additional tables, fields, or functionalities can be added easily in the future without affecting the overall database structure significantly.

Overall, the relational schema provides a clear and organized framework for designing and managing the database. It contributes greatly to improving data organization, operational efficiency, and reliability within the Pharmacy Management System.

CHAPTER 4

FUNDAMENTALS OF SQL

4.1 Introduction

Structured Query Language (SQL) is a standard language used to communicate with databases. It helps users create, store, manage, update, and retrieve data from a database efficiently. SQL plays an important role in Database Management Systems because it provides simple and organized commands for handling large amounts of data systematically.

In database-driven applications, SQL is used to perform various operations such as inserting new records, updating existing information, deleting unwanted data, and retrieving required records from the database. These operations help maintain data accurately, securely, and efficiently within the system.

In the Pharmacy Management System, SQL is used to manage medicine details, customer information, supplier records, inventory data, sales history, and billing information stored in the SQLite database. SQL queries help the application interact with the database smoothly and allow users to access, update, and manage records whenever required.

One of the major advantages of SQL is that it is easy to understand and use. SQL commands are simple and resemble normal English statements, making database operations easier for developers and users. Commands such as `SELECT`, `INSERT`, `UPDATE`, and `DELETE` help perform different database operations efficiently.

SQL also supports data security, consistency, and efficient data management. It helps ensure that records are stored properly and retrieved accurately whenever needed. In applications like the Pharmacy Management System, maintaining accurate data is important for handling billing, medicine stock, and customer records correctly.

Another important feature of SQL is faster data retrieval. Users can search and access required information quickly without manually checking records. This improves the speed and efficiency of pharmacy operations and helps save time during billing and inventory management.

SQL also helps reduce data redundancy by organizing records systematically within tables. Proper table relationships and structured queries improve database performance and make data management more organized.

In this project, SQL queries are used for adding new medicine records, updating stock details, retrieving customer information, generating billing records, and managing inventory efficiently. SQL works together with Python and Flask to provide smooth communication between the application and the SQLite database.

Another advantage of SQL is its reliability and scalability. Large amounts of pharmacy-related data can be managed easily without affecting system performance. Additional records and features can also be added in the future with minimal difficulty.

SQL is widely used in many real-world applications because it helps organize, process, and manage data systematically. It allows databases to handle multiple records efficiently and improves the overall performance, reliability, and usability of applications.

Overall, SQL plays a very important role in the Pharmacy Management System by supporting efficient storage, retrieval, updating, and management of pharmacy-related information. It helps make the application more organized, reliable, accurate, and efficient for daily pharmacy operations.

4.2 SQL Commands

SQL provides different types of commands that are used to perform various operations on a database. These commands help users create database structures, store and manage records, control access permissions, and retrieve required information efficiently. SQL commands are grouped into different categories based on the type of operation they perform.

One important category is Data Definition Language (DDL). DDL commands are used to define and manage the structure of database objects such as tables and schemas. Commands like CREATE, ALTER, and DROP are included in this category. They help in creating new tables, modifying existing structures, and deleting unwanted database objects.

Another category is Data Manipulation Language (DML). DML commands are used to insert, update, and delete records stored in the database. Commands such as INSERT, UPDATE, and DELETE help in managing the data within tables. In the Pharmacy Management System, DML commands are used for adding medicine details, updating inventory records, and managing billing information.

Data Control Language (DCL) commands are used for controlling user permissions and access to the database. Commands like GRANT and REVOKE help in providing or removing access rights for different users. These commands improve database security and protect important information from unauthorized access.

Transaction Control Language (TCL) commands are used to manage database transactions. Commands such as COMMIT, ROLLBACK, and SAVEPOINT help in maintaining data consistency and ensure that database operations are completed successfully without errors.

Data Query Language (DQL) is mainly used for retrieving information from the database. The SELECT command is the most commonly used DQL command and helps users fetch required data from tables easily.

In the Pharmacy Management System, these SQL commands are used for creating tables, storing medicine and customer records, updating inventory details, generating billing information, and retrieving data whenever required. They help in managing the database efficiently and ensure smooth functioning of the application.

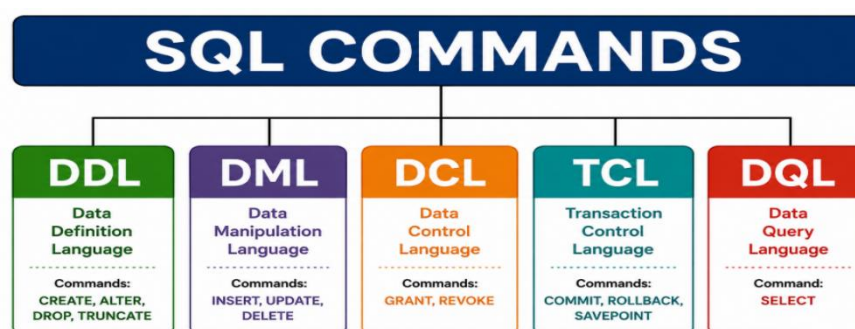


Figure 4.1 – Types of SQL Commands

4.3 Data Definition Language (DDL)

Data Definition Language (DDL) is a category of SQL commands used to define and manage the structure of database objects. These commands help in creating, modifying, and deleting tables and other database components. DDL plays an important role in designing and organizing the database properly before storing data.

One of the most commonly used DDL commands is CREATE. The CREATE command is used to create new database objects such as tables, views, and indexes. In the Pharmacy Management System, the CREATE command is used to create tables for storing medicine details, customer information, inventory records, and billing data.

Another important DDL command is ALTER. The ALTER command is used to modify the structure of an existing table. It allows users to add new columns, remove columns, or

change the data type of existing fields whenever necessary. This command is useful when updates or improvements need to be made to the database structure.

The DROP command is used to permanently delete database objects such as tables or entire databases. Once a table is dropped, all the data stored inside it is removed completely. Therefore, this command should be used carefully.

Apart from these, DDL also includes commands like TRUNCATE and RENAME. The TRUNCATE command removes all records from a table without deleting the table structure, while the RENAME command is used to change the name of a database object.

In the Pharmacy Management System, DDL commands help in creating and maintaining the database structure efficiently. They ensure that the database is organized properly and supports smooth storage and management of pharmacy-related information.

Category	Commands	Purpose
DDL	CREATE, ALTER	Create Tables
DML	INSERT, UPDATE	INSERT, UPDATE
DQL	SELECT	Retrieve Data
TCL	COMMIT	Save Transactions

Table 4.1 – SQL Command Categories

4.4 Data Manipulation Language (DML)

Data Manipulation Language (DML) is a category of SQL commands used to manage and manipulate the data stored in a database. These commands help users insert new records, update existing information, and delete unwanted data from database tables. DML commands are widely used in database-driven applications because they allow smooth interaction with stored data.

One of the most commonly used DML commands is INSERT. The INSERT command is used to add new records into a table. In the Pharmacy Management System, this

command is used for storing new medicine details, customer information, billing records, and inventory data into the database.

Another important DML command is UPDATE. The UPDATE command is used to modify or change existing records in a table. For example, in this project, it can be used to update medicine stock details, change medicine prices, or modify customer information whenever necessary.

The DELETE command is used to remove records from a table. It helps in deleting unwanted or outdated information from the database. In the Pharmacy Management System, DELETE operations may be used for removing expired medicine records or unnecessary data entries.

DML commands help in maintaining the database dynamically because the stored data can be managed and modified whenever required. These commands improve the flexibility and efficiency of the database system and make data handling easier for users.

In this project, DML commands play an important role in managing pharmacy-related information such as medicine records, inventory details, customer data, and billing information. They help in keeping the database updated and organized throughout the operation of the system.

4.5 Data Control Language (DCL)

Data Control Language (DCL) is a category of SQL commands used to control access and permissions in a database system. These commands help in managing the security of the database by deciding which users are allowed to access, modify, or manage the stored data. DCL plays an important role in protecting sensitive information and maintaining overall database security.

The two main DCL commands are GRANT and REVOKE.

The GRANT command is used to provide specific permissions or access rights to users. These permissions may include the ability to view data, insert new records, update information, or delete records from database tables. By using the GRANT command, database administrators can control which users are allowed to perform certain operations within the system.

The REVOKE command is used to remove permissions that were previously granted to users. It helps restrict unauthorized access and ensures that only approved users can perform operations on the database. If a user no longer requires access to certain records or functionalities, the REVOKE command can be used to remove those permissions safely.

In applications like the Pharmacy Management System, database security is very important because the system stores sensitive information such as customer details, billing records, medicine stock, supplier data, and inventory information. DCL commands help protect this information from unauthorized access, accidental modification, or misuse.

Another important advantage of DCL is controlled user management. Different users can be given different levels of access depending on their roles and responsibilities. For example, an administrator may have full access to the database, while pharmacy staff may only be allowed to view inventory details or generate bills.

DCL also helps improve data privacy and confidentiality by limiting access to important records. This ensures that sensitive pharmacy data is accessed only by authorized individuals, improving the reliability and security of the system.

By controlling database permissions properly, DCL helps prevent unauthorized changes that may affect the accuracy and consistency of records. This is especially important in applications where billing information and medicine records must remain accurate at all times.

DCL commands also support better database management and system monitoring because user activities and access rights can be controlled systematically. This improves overall organization and security within the application.

In the Pharmacy Management System, DCL contributes to secure handling of pharmacy operations by protecting medicine records, customer information, billing transactions, and inventory data from unauthorized usage.

Overall, Data Control Language plays an important role in maintaining database security, access control, and data protection. It helps make the Pharmacy Management System more secure, reliable, and organized by ensuring that only authorized users can access and manage important pharmacy-related information.

Data Control Language improves the overall safety and reliability of the database system. It helps administrators manage user access efficiently and ensures proper control over the stored information.

Although SQLite provides limited support for advanced permission management compared to larger database systems, the concept of DCL remains important in understanding database security and access control in real-world applications.

4.6 Transaction Control Language (TCL)

Transaction Control Language (TCL) is a category of SQL commands used to manage transactions in a database system. A transaction is a group of database operations performed together as a single unit of work. TCL commands help ensure that these operations are completed accurately, securely, and consistently within the database.

TCL is mainly used to maintain data integrity and avoid errors during database operations. If multiple operations are being performed together, TCL ensures that either all operations are completed successfully or none of them are applied. This helps prevent incomplete updates, incorrect records, and data inconsistency within the system.

The most commonly used TCL commands are COMMIT, ROLLBACK, and SAVEPOINT.

The COMMIT command is used to permanently save the changes made to the database. Once the changes are committed, they become permanent and cannot be undone easily. In the Pharmacy Management System, the COMMIT command can be used after successfully generating a bill and updating medicine stock records.

The ROLLBACK command is used to undo changes made during a transaction if any error occurs. This helps restore the database to its previous stable state and prevents incorrect or incomplete data from being stored. For example, if a billing operation fails while updating inventory, the rollback process helps cancel incomplete changes and maintain accurate records.

The SAVEPOINT command is used to create temporary points within a transaction. It allows partial rollback to a specific point instead of undoing the entire transaction. This provides better control over complex database operations and improves flexibility during transaction handling.

Another important advantage of TCL is improved reliability of database operations. Since transactions are managed carefully, the chances of data corruption and inconsistent updates are reduced significantly. This is especially important in applications that handle financial transactions and inventory updates.

In the Pharmacy Management System, TCL commands help maintain accurate billing records, inventory updates, customer transactions, and medicine stock management. For example, while generating a bill and reducing medicine quantity from inventory together, TCL ensures that all related operations are completed properly without data inconsistency.

TCL also improves database stability by handling unexpected failures or interruptions effectively. If a system error occurs during a transaction, rollback operations help protect the database from storing incomplete information.

By managing transactions properly, TCL contributes to better data accuracy, consistency, and reliability within the application. It ensures that pharmacy records remain correct and properly synchronized after every operation.

Overall, Transaction Control Language plays an important role in maintaining reliable and consistent database operations. In the Pharmacy Management System, TCL helps ensure secure transaction handling, accurate record management, and smooth execution of important pharmacy-related operations.

4.7 Data Query Language (DQL)

Data Query Language (DQL) is a category of SQL commands used to retrieve and display data from a database. DQL helps users access the required information stored in database tables quickly, accurately, and efficiently. It is one of the most frequently used parts of SQL because most database operations involve viewing, searching, and analyzing stored data.

The main command used in DQL is the SELECT statement. The SELECT command is used to fetch records from one or more tables based on user requirements. It allows users to view specific data instead of displaying the entire database contents, which improves efficiency and reduces unnecessary data processing.

The SELECT statement can also be combined with conditions and clauses such as WHERE, ORDER BY, GROUP BY, and HAVING to filter, sort, and organize the retrieved data. These clauses make data retrieval more flexible and help users access information in a structured and meaningful manner.

For example, the WHERE clause can be used to display medicines with low stock quantity, while the ORDER BY clause can sort medicine records alphabetically or based on price. Such features help improve data management and reporting within the application.

In the Pharmacy Management System, DQL commands are used to retrieve medicine details, customer information, supplier records, billing history, inventory data, and sales reports from the database. The SELECT statement can be used to check available medicine stock, search customer records, display recent transactions, or retrieve billing details whenever required.

DQL plays an important role in making database systems user-friendly because it allows users to access information easily without modifying the stored data. Pharmacy staff can quickly search and retrieve records without manually checking files or registers, which saves time and improves operational efficiency.

Another advantage of DQL is faster data retrieval. Even when the database contains a large number of records, SQL queries help users access required information quickly and accurately. This improves application performance and supports smooth workflow management.

DQL is also useful for generating reports and analyzing stored information. In the Pharmacy Management System, DQL queries can help display sales reports, inventory summaries, customer purchase history, and medicine availability records in an organized format.

The use of DQL improves decision-making because users can retrieve updated and accurate information whenever required. This helps pharmacy staff manage stock levels, monitor billing activities, and maintain customer records more effectively.

Overall, Data Query Language plays a major role in improving data accessibility, organization, and retrieval efficiency within the database system. In the Pharmacy Management System, DQL helps retrieve pharmacy-related information quickly, accurately, and systematically, making the application more reliable, interactive, and efficient.

CHAPTER 5

DESIGN

5.1 Design Goals

The main goal of designing the Pharmacy Management System is to create an application that is simple, efficient, reliable, and easy to use. The system is designed in such a way that users can perform pharmacy-related operations smoothly without facing technical difficulty. Special attention is given to creating a user-friendly interface so that even users with basic computer knowledge can easily access and operate the application.

Another important design goal of the project is efficient database management. The system is designed to store and manage medicine details, customer records, supplier information, inventory data, sales history, and billing records in an organized manner. Proper database design helps reduce data redundancy, improve data accuracy, and ensure faster retrieval of information whenever required.

The project also focuses on reducing manual work and minimizing human errors in pharmacy management. Features such as automated billing, inventory management, digital record maintenance, and organized transaction handling help improve the overall efficiency of pharmacy operations.

Speed and accuracy are also important goals of the system design. The application is developed to perform operations such as billing generation, stock updates, and data retrieval quickly and accurately. This helps pharmacy staff complete daily tasks more efficiently and improves workflow management.

Security and reliability are also considered important during the design process. The system is designed to maintain records safely and ensure that important information is stored properly without data loss or unauthorized access. Secure handling of customer details, billing records, and medicine information improves system reliability.

Another objective of the design is to make the application scalable and maintainable. The modular structure of the project allows future improvements, additional modules, and advanced features to be added easily whenever required. This flexibility helps support future expansion of the system.

The design also focuses on better user interaction and accessibility. Clear navigation menus, dashboard options, organized forms, and responsive page layouts help users interact with the application comfortably and perform operations without confusion.

Integration between frontend, backend, and database modules is another important design goal. Smooth communication between the user interface, Flask backend, Python functions, and SQLite database helps ensure efficient functioning of the complete system.

The project is also designed to improve organization of pharmacy records by maintaining all information digitally instead of using manual registers or paperwork. This helps reduce record mismanagement and improves overall productivity.

Another important goal is maintaining consistency and reliability of data. Proper database structure and transaction management help ensure that records remain accurate and synchronized during billing, inventory updates, and customer management operations.

Overall, the design goals of the Pharmacy Management System focus on simplicity, efficiency, speed, accuracy, security, scalability, and better management of pharmacy-related activities through a well-organized and user-friendly digital solution.

Module	Function
Inventory Management	Manage medicine stock
Billing System	Generate bills
Customer Management	Maintain customer records
Supplier Management	Store supplier details
Sales Management	Manage sales records
Prescription Module	Upload prescriptions

Table 5.1 – Modules of Pharmacy Management System

5.2 Database Structure

The database structure of the Pharmacy Management System is designed to store and manage pharmacy-related information in an organized, secure, and efficient manner. SQLite is used as the Database Management System because it is lightweight, simple to use, and suitable for small and medium-scale applications. The database helps maintain records digitally and supports smooth interaction between the application and stored data.

The database consists of different tables that are used to store various types of information such as medicine details, customer records, supplier information, inventory data, sales history, and billing records. Each table is designed with specific fields and attributes to organize the data properly and systematically.

The medicine table stores details related to medicines such as medicine ID, medicine name, category, quantity, price, manufacturer details, supplier information, and expiry date. This table helps manage stock availability and maintain inventory records efficiently.

The customer table contains customer-related information such as customer ID, customer name, phone number, email address, and billing details. Maintaining customer records digitally helps improve service management and transaction tracking within the pharmacy.

Another important table in the database is the billing table, which stores billing details such as bill number, purchased medicines, quantity, total amount, GST details, payment information, and transaction date. These records help maintain proper billing history and support accurate financial management.

The database may also contain supplier and inventory tables to manage supplier details, medicine stock updates, stock intake, and returns management. These tables help improve overall pharmacy organization and inventory handling.

Relationships between tables are maintained using primary keys and foreign keys to ensure proper connection between related data. This helps reduce data redundancy and improves consistency and accuracy within the database.

The database structure is designed in a way that allows easy insertion, updating, deletion, and retrieval of records. SQL queries are used to perform these operations efficiently and support smooth functioning of different modules within the application.

Another important advantage of the database structure is faster data retrieval. Pharmacy staff can quickly search for medicine details, billing records, inventory information, and customer data whenever required. This improves workflow efficiency and saves time during daily pharmacy operations.

The structured database design also improves scalability and maintainability of the application. Additional tables, fields, or features can be added in the future without affecting the overall system significantly.

Data security and proper organization are also improved through the database structure. Records are stored systematically, which reduces the chances of data duplication, loss, or incorrect updates.

The SQLite database is the backbone of the Pharmacy Management System — keeping inventory, billing, customers, suppliers, and transactions all running smoothly in one place. It ensures records stay accurate, data stays organized, and the whole system just works.

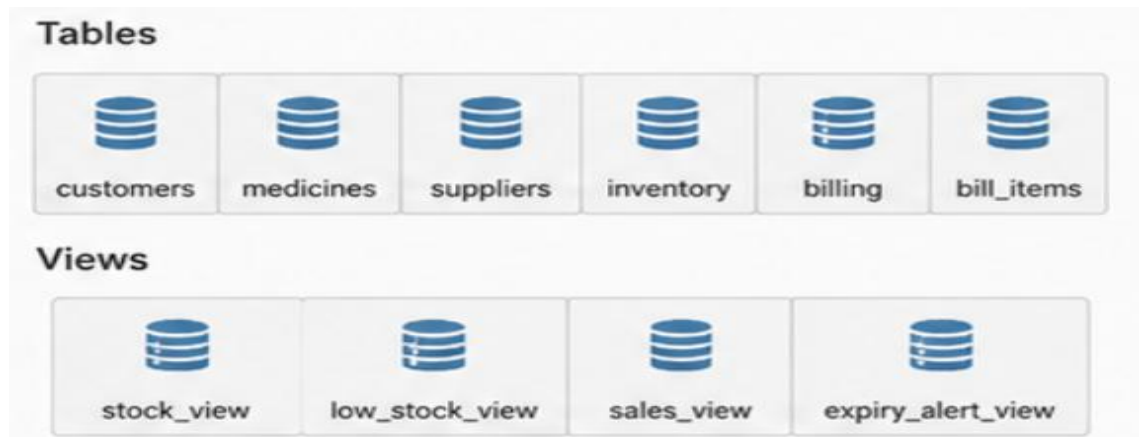


Figure 5.2 – Various Tables and Views in the Database

Column	Data Type	Description
cust_id	INTEGER	Customer ID
cust_name	TEXT	Customer Name
phone	TEXT	Contact Number
email	TEXT	Email address
address	TEXT	Customer address

Table 5.2 – Customer Table Structure

Field Name	Data Type	Description
medicine_id	INTEGER	Unique medicine ID
medicine_name	TEXT	Name of medicine
quantity	INTEGER	Available stock
price	Float	Medicine price
expiry_date	Date	Expiry date

Table 5.3 – Medicine Table Structure

5.3 GUI Structure

The Graphical User Interface (GUI) of the Pharmacy Management System is designed to provide a simple, interactive, and user-friendly experience for users. The interface helps users perform pharmacy-related operations easily without needing advanced technical knowledge. The GUI is developed using HTML templates and CSS styling, which help in creating structured, responsive, and visually organized web pages for the application.

The application consists of different pages and modules that handle various pharmacy operations efficiently. The dashboard acts as the main page of the system and provides easy navigation to different sections of the application. It helps users access important features such as inventory management, customer records, billing, supplier details, sales information, and medicine stock updates quickly and efficiently.

The inventory management page is used for adding, updating, viewing, and managing medicine stock details. It allows users to check medicine availability, update quantities, monitor stock levels, and maintain inventory records properly.

The billing page helps generate bills and maintain transaction details. Users can enter medicine information, calculate total amounts, apply taxes or discounts, and store billing records efficiently through this module. The billing interface is designed to make transaction handling faster and more accurate.

The customer management page allows users to store and manage customer details such as names, contact information, addresses, and purchase history. This helps improve customer service and maintain organized records within the pharmacy.

The GUI is designed in a clean, simple, and organized manner so that users can interact with the system comfortably. Proper layout design, navigation menus, buttons, tables, and form structures improve usability and make the application easier to understand and operate.

Another important feature of the GUI structure is smooth navigation between modules. Users can move from one section to another easily without confusion, which improves workflow efficiency during pharmacy operations.

HTML templates also help separate the frontend design from the backend logic, making the application more organized, maintainable, and scalable. Flask is used to connect the GUI with backend operations and database functionality efficiently.

Additional styling and formatting improve the appearance and user experience of the application. Proper alignment, spacing, colors, and responsive page design make the interface more attractive and user-friendly.

The GUI also improves accessibility by presenting information clearly in tables, forms, and dashboard sections. Medicine details, billing records, inventory updates, and customer information can be viewed and managed easily through the interface.

In the Pharmacy Management System, the GUI plays a major role in simplifying user interaction and improving the overall usability of the application. A well-designed interface helps pharmacy staff perform operations quickly, accurately, and comfortably.

Overall, the GUI structure of the Pharmacy Management System focuses on simplicity, accessibility, responsiveness, and smooth interaction between the user and the application while improving efficiency and user experience.

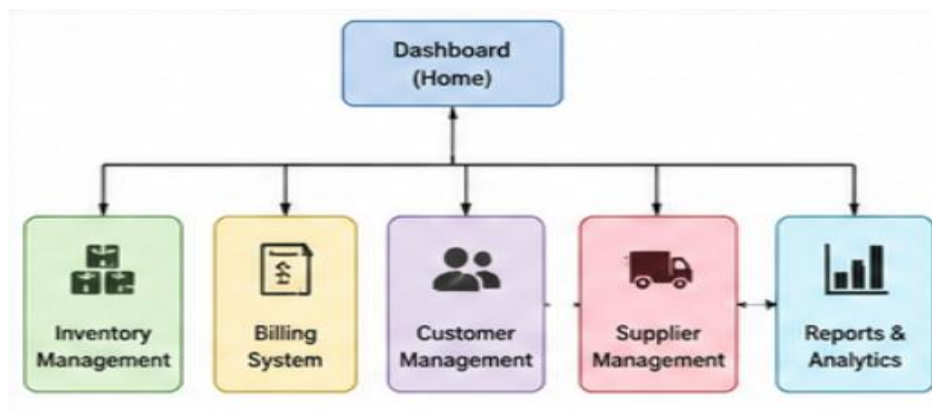


Figure 5.1 – Pharmacy Dashboard Layout

CHAPTER 6

IMPLEMENTATION

6.1 Creating the Database

The database for the Pharmacy Management System is created using SQLite, which is a lightweight and efficient database management system. SQLite is chosen for this project because it is simple to use, requires minimal setup, and is suitable for small and medium-scale applications. It stores the entire database in a single file, making it easier to manage and maintain.

During the database creation process, different tables are designed to store various types of pharmacy-related information. These tables help in organizing data systematically and allow smooth interaction between the application and the database.

Separate tables are created for storing medicine details, customer information, inventory records, and billing data. Each table contains specific fields and attributes based on the type of information being stored. For example, the medicine table may include fields such as medicine ID, medicine name, quantity, price, and expiry date.

Primary keys and relationships are also defined while creating the database tables to ensure proper organization and consistency of data. This helps in avoiding duplicate records and improves data management within the system.

SQL commands such as CREATE TABLE are used to create the required tables in SQLite. These commands define the structure of the tables and the data types of different fields. The database is then connected to the application using Python so that records can be inserted, updated, deleted, and retrieved whenever required.

The database creation process plays an important role in the Pharmacy Management System because it provides the foundation for storing and managing all pharmacy-related information efficiently and securely.

6.2 Connecting the Database to the Application

The Pharmacy Management System connects Python with the SQLite database to store, retrieve, and manage pharmacy-related information efficiently. Database connectivity is an important part of the project because it allows the application to interact with the stored records and perform various operations smoothly.

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Python provides built-in database modules that make it easy to connect with SQLite databases. In this project, the SQLite database module is used to establish the connection between the application and the database file. Once the connection is created, the application can execute SQL queries for inserting, updating, deleting, and retrieving records.

The connection process begins by importing the required SQLite module in Python. After that, a connection object is created, which links the Python application to the SQLite database. Using this connection, different database operations are performed whenever users interact with the system.

For example, when a user adds a new medicine or generates a bill, the application sends SQL commands through Python to store the information in the database. Similarly, when medicine details or customer records need to be viewed, Python retrieves the required data from the SQLite database and displays it in the application interface.

Database connectivity helps in maintaining accurate and organized records within the system. It also improves the efficiency of the application by enabling faster data storage and retrieval operations.

Overall, connecting Python with SQLite plays a major role in the proper functioning of the Pharmacy Management System and helps in managing pharmacy data efficiently and securely.

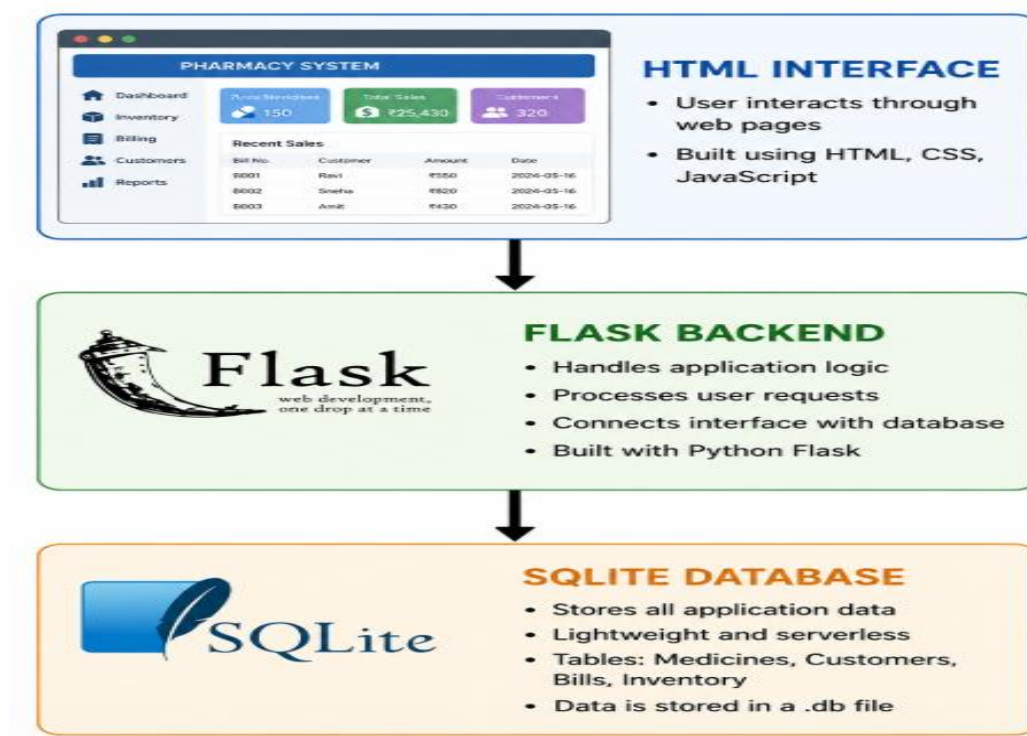


Figure 6.1 – Flask Application Connected to SQLite

6.3 Creating the Main Window

The main window of the Pharmacy Management System acts as the central interface of the application and provides users with access to different features and modules. It is designed in a simple and user-friendly manner so that users can navigate through the system easily without confusion.

In this project, the dashboard functions as the main window of the application. It serves as the homepage that appears after the user accesses the system. The dashboard contains navigation options and links to different sections such as inventory management, customer records, medicine details, and billing operations.

The main window is designed using HTML templates to provide a clean and organized interface. Proper layout design and navigation menus help users move between different modules smoothly. The dashboard also displays important information and options in a structured way, making the application easier to use.

The purpose of the main window is to provide quick access to all major functionalities of the Pharmacy Management System from a single interface. Instead of opening multiple applications or pages separately, users can perform different operations directly through the dashboard.

The dashboard improves the overall user experience by making the application more interactive and accessible. It also helps in reducing complexity and ensures smooth communication between the frontend interface and backend operations.

Overall, the main window plays an important role in the Pharmacy Management System by acting as the central control interface for managing pharmacy-related activities efficiently.

6.4 Displaying Frames Over the Main Window

The Pharmacy Management System is divided into different modules to make the application more organized and easier to use. Each module performs a specific function such as inventory management, billing, customer management, or medicine record handling. These modules are displayed through separate pages and interfaces within the application.

The main dashboard acts as the central window from which users can access different sections of the system. When a user selects a particular option from the dashboard, the corresponding page or frame is displayed. This approach helps in separating functionalities clearly and improves the overall structure of the application.

Separate pages are designed for different operations such as adding medicines, viewing inventory details, managing customer records, and generating bills. Each page contains forms, buttons, and fields related to the specific task being performed. This makes the application more user-friendly and reduces confusion while navigating through different features.

The frontend interface is created using HTML templates, which help in displaying the pages in a structured and visually organized manner. Flask framework manages the navigation between these pages and ensures smooth communication between the frontend and backend of the application.

Displaying different modules through separate pages also improves maintainability of the project. If any modifications or updates are required in a particular module, they can be made easily without affecting the entire application.

Overall, dividing the application into separate frames and pages improves organization, usability, and efficiency of the Pharmacy Management System and provides a better experience for users.

6.5 Processing Queries

Processing queries is an important part of the Pharmacy Management System because it allows the application to interact with the database and manage stored information efficiently. SQL queries are used to perform various database operations such as inserting new records, updating existing data, deleting unwanted information, and retrieving records whenever required.

In this project, SQL queries are executed using Python code through SQLite database connectivity. Python acts as a bridge between the application interface and the database by sending SQL commands and processing the returned results.

Whenever a user performs an action in the application, such as adding a new medicine, updating inventory details, generating a bill, or viewing customer records, the corresponding SQL query is executed automatically in the background. These queries help in storing and retrieving data accurately from the database.

Different types of SQL queries are used depending on the operation being performed. INSERT queries are used for adding new records, UPDATE queries are used for modifying existing information, DELETE queries remove unwanted records, and SELECT queries retrieve data from the database.

The Python code processes these queries using database connection objects and cursor methods provided by the SQLite module. After executing a query, the changes are committed to the database to ensure that the records are saved properly.

Query processing improves the overall functionality and efficiency of the Pharmacy Management System by enabling smooth communication between the application and the database. It helps in maintaining accurate pharmacy records and ensures proper management of inventory, customer information, and billing data.

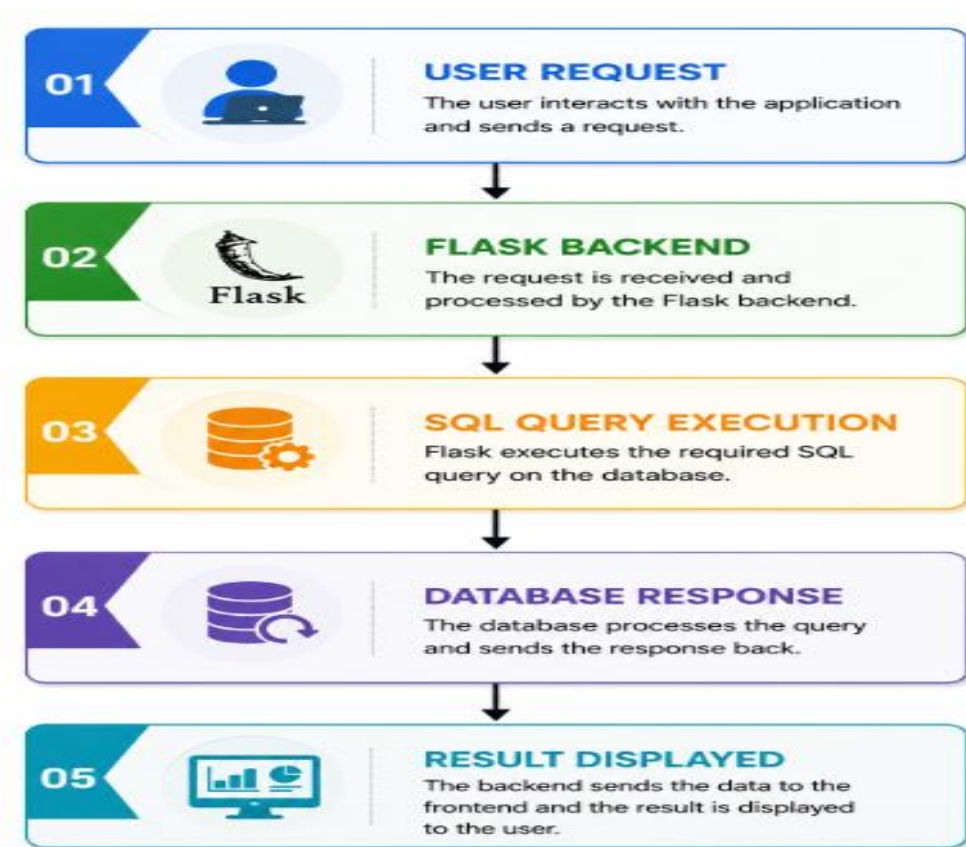


Figure 6.2 – Query Processing Workflow

6.6 System Functionality

The Pharmacy Management System is designed to perform different pharmacy-related operations efficiently through various integrated functionalities. The system helps in automating daily pharmacy activities and reduces the manual effort required for maintaining records and managing transactions.

One of the main functionalities of the system is inventory management. The application allows users to add new medicines, update stock details, check medicine availability,

monitor expiry dates, and manage inventory records efficiently. It helps pharmacists maintain accurate stock information and reduces the chances of medicine shortages, duplicate entries, or incorrect inventory handling.

Another important functionality is billing and sales management. The system helps in generating bills quickly and accurately by calculating medicine prices, quantities, taxes, and total amounts automatically. Billing records are stored in the database for future reference, which improves transaction management and reduces manual calculation errors. The sales module also helps users track recent transactions and maintain proper sales history.

The system also includes customer management functionality. Customer details such as names, contact information, addresses, and purchase records can be stored and managed efficiently within the application. This helps in maintaining organized customer records, improving transaction tracking, and providing better customer service.

Supplier management is another useful functionality included in the system. Pharmacy staff can maintain supplier details, manage medicine purchases, and track supplier-related records easily. This improves overall stock management and ensures smooth pharmacy operations.

Apart from these features, the application also supports database management, data retrieval, stock intake management, returns handling, and record updating functionalities. Users can easily search for medicine details, view inventory information, retrieve billing records, and update pharmacy data whenever required.

The system is designed with a simple, interactive, and user-friendly interface so that pharmacy staff can perform operations smoothly without technical difficulty. Proper navigation menus and dashboard options help users access different modules quickly and efficiently.

Another important feature of the Pharmacy Management System is secure login authentication, which helps protect sensitive pharmacy records and restrict unauthorized access to the application. This improves data privacy and system reliability.

By automating important pharmacy tasks, the application improves efficiency, saves time, reduces paperwork, and minimizes human errors. It also helps in maintaining proper organization of pharmacy data and improves the overall workflow within the pharmacy.

Overall, the Pharmacy Management System provides an organized, reliable, and efficient solution for managing pharmacy operations digitally while improving productivity, accuracy, and ease of management.

CHAPTER 7

RESULTS

7.1 Login Page

The login page is one of the important components of the Pharmacy Management System because it provides secure access to the application. It acts as the entry point of the system and ensures that only authorized users can access the pharmacy records and functionalities.

The login interface is designed in a simple and user-friendly manner so that users can enter their credentials easily. Users are required to provide details such as username and password to access the system. Once the entered credentials are verified successfully, the user is redirected to the main dashboard of the application.

The main purpose of the login page is to maintain security and protect important pharmacy data from unauthorized access. Since the system stores sensitive information such as medicine details, billing records, inventory data, and customer information, secure authentication is necessary.

The login page also improves the organization of the system by allowing controlled access to different users. Proper validation is performed to check whether the entered login details are correct before granting access to the application.

In this project, the login functionality is connected with the backend and database using Python and Flask. The entered details are processed and verified through database queries to ensure secure authentication. Error messages are displayed when invalid credentials are entered, which helps users understand login issues clearly and improves the overall user experience.

The design of the login page is kept simple and clean so that users can navigate the application without confusion. It also helps in maintaining professionalism and improves the overall appearance of the Pharmacy Management System.

Overall, the login page plays an important role in maintaining the privacy, security, and reliability of the Pharmacy Management System while providing users with safe, smooth, and controlled access to the application.

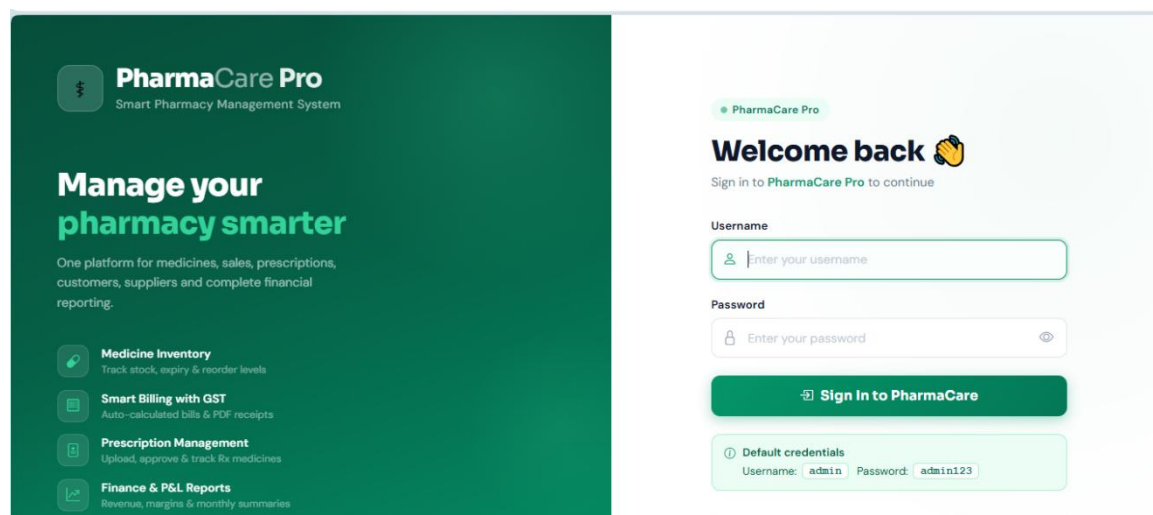


Figure 7.1 – Login Page

7.2 Dashboard

The dashboard is the main interface of the Pharmacy Management System and acts as the central control panel of the application. It provides users with quick access to different modules and features available in the system. After successful login, users are redirected to the dashboard, where they can navigate to various sections such as inventory management, billing, customer records, supplier details, sales management, and medicine information.

The dashboard is designed in a simple, clean, and organized manner to improve user experience and make navigation easier. It displays important pharmacy-related information clearly so that users can access required functionalities without confusion. Different buttons, menus, icons, and navigation options are provided to help users move between modules smoothly and efficiently.

The dashboard helps users manage daily pharmacy operations effectively by providing centralized access to all major functionalities from a single page. Instead of searching through multiple pages manually, users can directly access the required section through the dashboard interface, which saves time and improves productivity.

In the Pharmacy Management System, the dashboard may display useful details such as medicine stock status, low-stock alerts, billing options, sales records, inventory updates, customer information, and supplier management features. Important notifications and quick-access features help pharmacy staff monitor daily activities more efficiently.

The dashboard interface is developed using HTML, CSS, Flask, and Python to provide a responsive and user-friendly environment. Proper layout design, organized menus, and

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attractive navigation structure make the application more interactive and easy to understand even for non-technical users.

Another important advantage of the dashboard is that it improves overall workflow management within the pharmacy. Users can perform multiple operations such as updating medicine stock, generating bills, checking inventory, and managing records from a single platform without difficulty.

The dashboard also contributes to better data organization and faster decision-making by presenting information in a structured manner. Since all important modules are connected through the dashboard, users can easily monitor pharmacy operations and maintain records more efficiently.

Overall, the dashboard plays an important role in improving the usability, organization, accessibility, and efficiency of the Pharmacy Management System by acting as the primary navigation and management interface of the application.

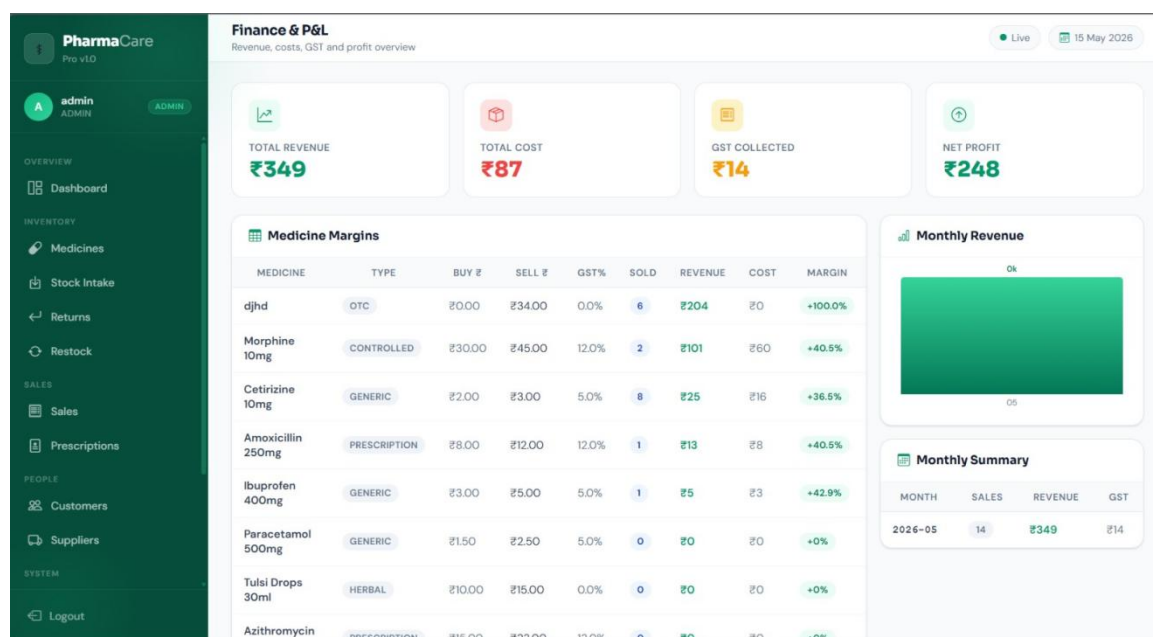


Figure 7.2 – Dashboard

7.3 Inventory Management

The inventory management module is one of the most important features of the Pharmacy Management System. It helps in maintaining and managing medicine stock efficiently within the pharmacy. Proper inventory management is essential to ensure

that medicines are available whenever required and to avoid problems such as stock shortages, expired medicines, or overstocking.

The inventory module allows users to add new medicine details, update existing stock information, and view the availability of medicines easily. Information such as medicine name, quantity, price, manufacturer details, category, supplier information, and expiry date can be stored and managed through this module.

Whenever medicines are sold or new stock is added, the inventory records are updated automatically in the database. This helps in maintaining accurate stock information and reduces the chances of manual errors in inventory handling. Automatic stock updates also help pharmacy staff monitor medicine availability more effectively.

The module helps pharmacists keep track of available medicines and monitor stock levels efficiently. Users can quickly search for specific medicines and check their availability whenever required. This improves the speed, accuracy, and efficiency of pharmacy operations.

Another useful feature of the inventory management module is stock monitoring and low-stock management. The system can help users identify medicines that are running low so that new stock can be arranged on time. This prevents interruptions in medicine availability and improves pharmacy service management.

The inventory management page is designed using HTML templates and connected with the SQLite database through Python and Flask. SQL queries are used to insert, update, retrieve, and manage inventory records within the database efficiently.

The module also supports proper organization of medicine details by categorizing records systematically. This makes stock handling easier and helps users maintain pharmacy records in a structured manner. Medicine information can be accessed quickly without manually searching through files or registers.

By automating inventory management, the system reduces paperwork and makes stock handling more organized, reliable, and efficient. It also helps in improving productivity, minimizing human errors, and maintaining proper pharmacy records digitally.

The inventory module plays an important role in improving the overall workflow of the pharmacy by simplifying stock management tasks and reducing the workload of pharmacy staff. It helps maintain better control over medicine availability and inventory updates.

Overall, the inventory management module ensures smooth and effective management of medicine stock within the Pharmacy Management System while improving accuracy, organization, and operational efficiency.

#	NAME	TYPE	QUANTITY	REORDER LEVEL	PRICE	GST %	EXPIRY	SUPPLIER	STATUS	ACTIONS
2	Amoxicillin 250mg	RX	36	15	₹12.00	12.0%	2026-12-01	PharmaCorp	OK	
8	Ashwagandha 60tab	HERBAL	100	10	₹18.00	0.0%	2028-05-05	Herballife	OK	
7	Azithromycin 500mg	RX	12	10	₹22.00	12.0%	2026-10-01	PharmaCorp	OK	
5	Cetirizine 10mg	OTC	20	15	₹3.00	5.0%	2026-11-22	AllergyPlus	OK	
3	Ibuprofen 400mg	OTC	4	20	₹5.00	5.0%	2026-09-30	MedSupply Co.	LOW	
6	Morphine 10mg	CONTROLLED	18	5	₹45.00	12.0%	2026-08-15	PainCare Ltd.	OK	
1	Paracetamol 500mg	OTC	150	20	₹2.50	5.0%	2026-06-15	MedSupply Co.	OK	
4	Tulsi Drops 30ml	HERBAL	60	10	₹15.00	0.0%	2026-03-10	Herballife	EXPIRED	
10	aaa	RX	5	10	₹555.00	0.0%	2029-05-05	aaaa	LOW	
9	djhd	OTC	14	10	₹34.00	0.0%	2026-06-06	dsd	OK	

Figure 7.3 Inventory Management

7.4 Customer Management

The customer management module is designed to store and manage customer-related information efficiently within the Pharmacy Management System. Maintaining customer records digitally helps in improving organization and makes it easier to access information whenever required during pharmacy operations.

The system allows users to add, update, view, and manage customer details such as customer name, contact number, address, email ID, and purchase information. These records are stored securely in the SQLite database and can be retrieved quickly through the application whenever needed.

Customer management helps pharmacists maintain proper transaction history and improves communication and service management. Instead of maintaining customer records manually in registers or files, the application stores all information digitally, reducing paperwork and minimizing the chances of data loss or incorrect record handling.

The module also makes searching and updating customer information easier and faster. Users can quickly retrieve customer details during billing or while checking previous purchase records. This improves efficiency, reduces waiting time, and helps pharmacy staff provide better customer service.

The customer management interface is designed using HTML templates and connected with the backend through Python and Flask. SQL queries are used to store, update, search, and retrieve customer records from the database efficiently.

Another important advantage of this module is that it helps in maintaining organized purchase history and billing information for customers. This can be useful for repeat customers who regularly purchase medicines from the pharmacy. It also helps pharmacists track customer transactions more effectively and maintain proper billing records.

The module improves overall data organization by storing customer information systematically within the database. Since all records are maintained digitally, users can access information quickly without searching through physical files or registers. This saves time and improves the workflow of the pharmacy.

The customer management system also contributes to better workflow management within the pharmacy by simplifying customer-related operations and reducing manual effort. Proper handling of customer records improves reliability and supports smooth functioning of the entire pharmacy system.

In addition, the module helps in building better customer relationships by maintaining accurate records and ensuring quick service during medicine purchases and billing. Organized customer data also helps in handling future transactions more efficiently.

The customer management feature improves the professionalism and efficiency of the pharmacy by ensuring that all customer information is maintained securely and systematically within the application.

Overall, the customer management module helps in maintaining accurate customer records, improving service quality, reducing manual errors, and enhancing the overall efficiency, organization, and reliability of the Pharmacy Management System.

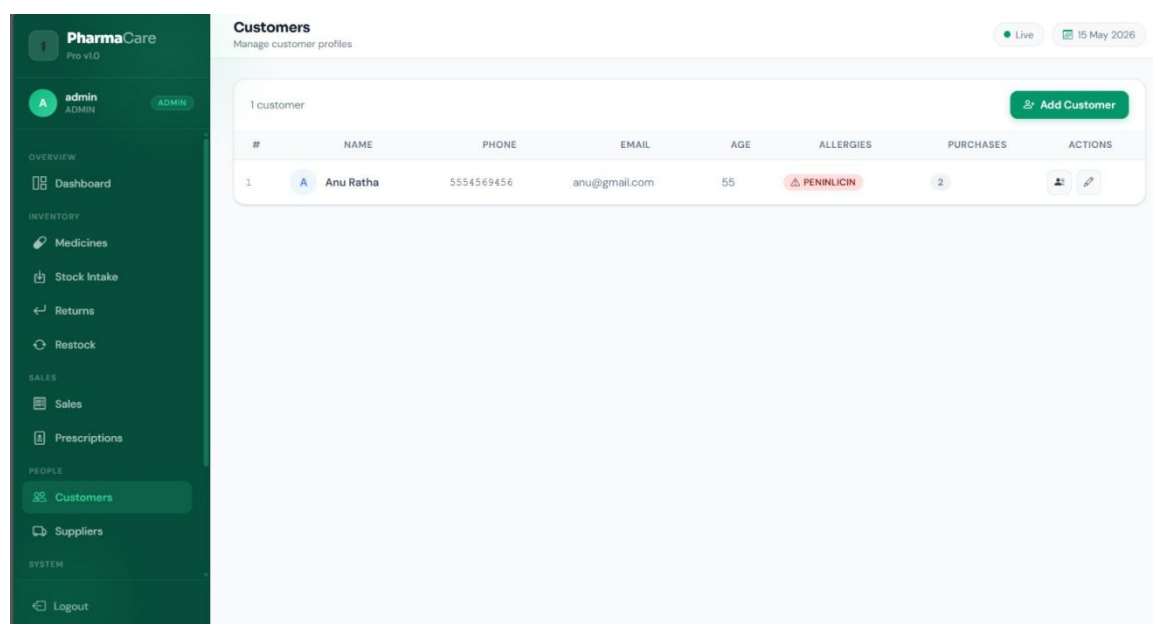


Figure 7.4 Customer Management

7.5 Billing System

The billing system is an important module of the Pharmacy Management System that helps in generating invoices and maintaining transaction records accurately. It automates the billing process and reduces the manual effort required for calculating medicine prices, quantities, taxes, discounts, and total amounts.

The billing module allows users to select medicines, enter quantities, and generate bills quickly and efficiently. The system automatically calculates the total amount based on the medicine price and quantity, which helps in reducing calculation errors and improving billing accuracy. This makes the billing process faster and more reliable during pharmacy operations.

All billing details are stored in the SQLite database for future reference. This helps in maintaining proper transaction history and allows users to retrieve previous billing records whenever required. Billing records may include details such as bill number, customer information, purchased medicines, quantity, payment method, GST details, and total payment amount.

The billing interface is designed in a simple, organized, and user-friendly manner so that pharmacy staff can generate invoices easily without technical difficulty. Proper buttons, forms, and navigation options make the billing process smooth and efficient for users.

The billing module is directly connected with the inventory management system, which helps in updating medicine stock automatically whenever a bill is generated. This ensures that inventory records remain accurate and updated after every transaction.

Another important feature of the billing system is sales record management. The application stores recent sales details and transaction history systematically, which helps pharmacy staff track daily sales and monitor pharmacy performance more effectively.

The billing system improves the speed and efficiency of pharmacy operations by reducing paperwork and minimizing manual calculations. It also helps in maintaining organized financial records and improves the overall management of sales transactions within the pharmacy.

The module supports better customer service by reducing billing time and helping staff complete transactions quickly. Customers can receive accurate bills without delays, which improves the overall experience and reliability of the pharmacy service.

The billing management system also helps pharmacy owners maintain proper financial tracking and transaction monitoring. Since all billing records are stored digitally, reports and transaction details can be accessed easily whenever required.

By automating billing operations, the system reduces human errors, improves calculation accuracy, and ensures secure handling of transaction data. It also contributes to better workflow management and improved operational efficiency within the pharmacy.

Overall, the billing module plays a major role in ensuring accurate transaction handling, organized record maintenance, faster sales processing, and providing a smooth billing experience for both pharmacy staff and customers within the Pharmacy Management System.

PHARMACY MANAGEMENT SYSTEM

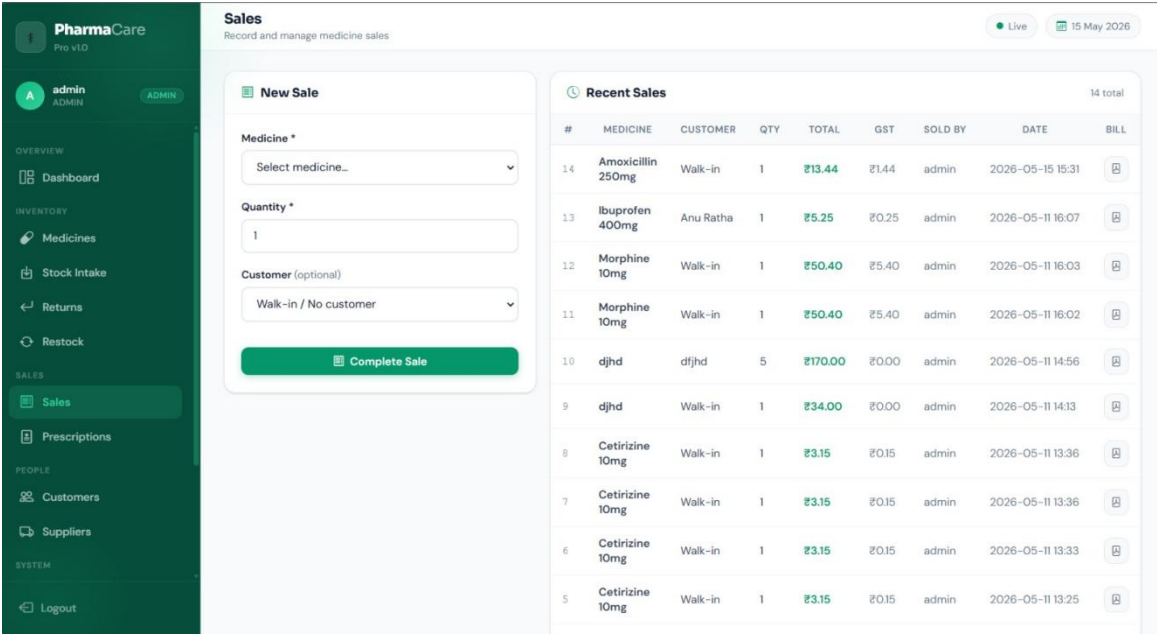


Figure 7.5 Billing System

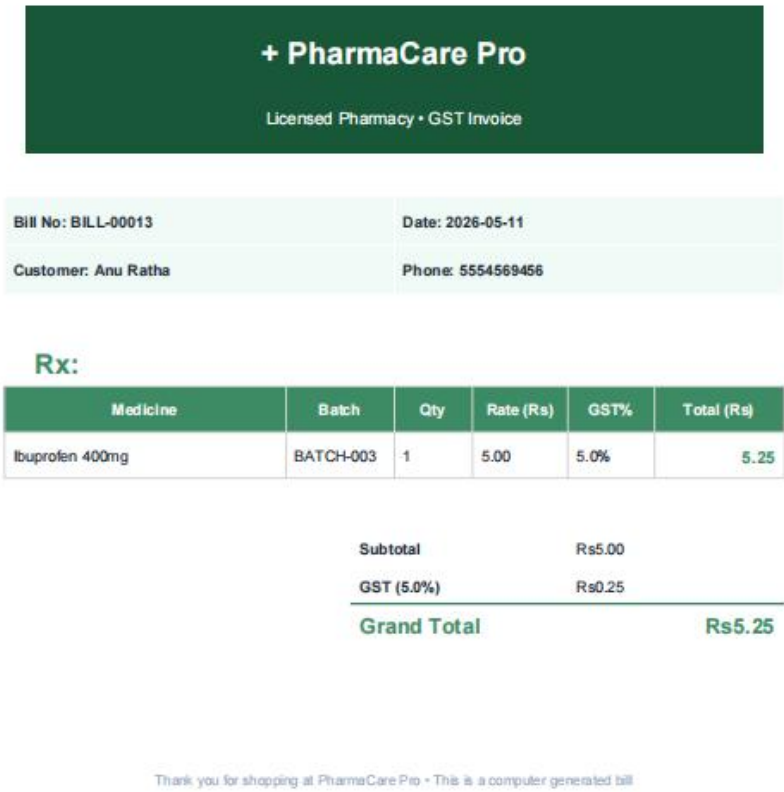


Figure 7.6 Customer Bill

7.5 USER UI (ORGANIZATION)

The User Interface (UI) of the Pharmacy Management System is designed in a simple, organized, and user-friendly manner to help users interact with the application easily. The interface focuses on providing smooth navigation and clear access to different modules such as inventory management, billing, customer records, supplier management, and sales details.

The UI is developed using HTML and CSS to create clean and responsive web pages. Proper layout design, navigation menus, buttons, tables, and forms are used to improve the overall appearance and usability of the system. The dashboard acts as the main interface of the application and helps users access important functionalities quickly without confusion.

Each module in the application is arranged systematically so that users can perform operations efficiently. For example, medicine details and stock records are displayed in an organized tabular format, which makes viewing and updating information easier. Billing pages are designed to simplify invoice generation and transaction handling.

The user interface also improves workflow management within the pharmacy by reducing the time required to access records and perform daily operations. Since the interface is designed to be simple and interactive, even users with basic computer knowledge can use the application comfortably.

Proper alignment, spacing, color combinations, and navigation structure help improve readability and user experience. The UI design also supports better communication between the frontend and backend modules through Flask integration.

Overall, the User Interface of the Pharmacy Management System helps improve accessibility, usability, and operational efficiency by providing an organized and interactive environment for managing pharmacy activities digitally.

7.6 LOGGING IN (ORGANIZATION)

The Login module is one of the important parts of the Pharmacy Management System because it provides secure and organized access to the application. It acts as the entry point of the system and ensures that only authorized users can access pharmacy records and system functionalities.

The login page is designed with a simple and user-friendly interface where users can enter their username and password easily. Once the entered details are verified successfully, the user is redirected to the dashboard of the application. This helps maintain proper organization and controlled access within the system.

The login system improves the security of pharmacy records by preventing unauthorized access to important information such as medicine details, customer records, billing data, and inventory information. Proper authentication ensures that sensitive pharmacy-related data remains protected and accessible only to approved users.

The login functionality is connected with the backend and SQLite database using Python and Flask. User credentials entered through the interface are verified through database queries before access is granted. This helps maintain reliability and improves the overall security of the application.

The organized login process also helps in maintaining proper workflow management within the pharmacy. Different users can access the system securely and continue daily operations such as billing, stock management, and customer handling efficiently.

The login page is designed to provide faster access and better user experience while maintaining simplicity and security. Error handling and validation mechanisms also help prevent incorrect login attempts and improve system reliability.

Overall, the Login module plays an important role in maintaining organization, privacy, security, and smooth functioning of the Pharmacy Management System by providing safe and controlled access to authorized users.

CHAPTER 8

CONCLUSION

8.1 Conclusion

The Pharmacy Management System is developed to simplify and improve the overall management of pharmacy operations through digital automation. The project successfully reduces manual work involved in maintaining medicine records, inventory details, customer information, and billing operations. By replacing traditional paper-based methods with a computerized system, the application helps in improving accuracy, efficiency, and organization within the pharmacy.

The system provides different functionalities such as inventory management, customer management, billing, supplier handling, and database management, which help pharmacy staff perform daily activities more easily and effectively. The application also ensures faster data retrieval, proper record maintenance, and better management of pharmacy-related information.

The use of Python, Flask, HTML, CSS, and SQLite helped in building a reliable, responsive, and user-friendly application. The project provided practical knowledge of web application development, database management, frontend designing, and backend integration. It also demonstrated how software solutions can be used to solve real-world management problems efficiently.

The Pharmacy Management System minimizes human errors, reduces paperwork, and improves productivity by automating important pharmacy tasks. Features such as medicine stock tracking, billing generation, sales management, prescription handling, and customer record maintenance make the system more organized and efficient for everyday pharmacy operations.

The system is designed in a simple and interactive manner so that users can access different features without technical difficulty. Proper navigation, secure login functionality, and systematic data storage improve the overall usability and reliability of the application.

Another important achievement of the project is improved inventory monitoring. Pharmacy staff can easily check medicine availability, monitor low stock conditions, and maintain proper medicine records. This helps in avoiding stock shortages and improves medicine management efficiency.

The project also helped in understanding the importance of database connectivity, secure authentication, and efficient data handling in modern management systems. It

provided valuable experience in integrating frontend and backend technologies to develop a complete working application.

Overall, the project was implemented successfully and achieved its objective of creating an efficient, organized, and reliable solution for pharmacy management. The system improves operational efficiency, saves time, and simplifies the handling of pharmacy activities through digital automation.

The Pharmacy Management System can be further enhanced in the future by adding advanced features such as online payment integration, medicine expiry notifications, report generation, barcode scanning, cloud-based storage, and mobile application support. These additional features can make the system more advanced, scalable, and suitable for larger pharmacy environments in the future.

8.2 Future Scope

The Pharmacy Management System can be further improved by adding several advanced features and functionalities in the future. Although the current system successfully manages pharmacy operations such as inventory handling, billing, customer management, supplier management, and sales tracking, additional enhancements can make the application more efficient, secure, scalable, and user-friendly.

One of the major future improvements can be the integration of online payment systems. This feature would allow customers to make digital payments through different payment methods such as UPI, debit cards, credit cards, net banking, or mobile wallets. Online payment integration would make the billing process faster, safer, and more convenient for both customers and pharmacy staff.

Another useful enhancement is the implementation of medicine expiry alerts and automatic stock notifications. The system can automatically identify medicines that are close to their expiry date and notify the pharmacy staff in advance. It can also generate low-stock alerts whenever medicine quantity falls below a certain limit. These features would help in preventing the sale of expired medicines and improve inventory management efficiency.

The application can also be extended by adding advanced report generation features. Different types of reports such as daily sales reports, monthly revenue reports, stock availability reports, supplier reports, GST reports, and customer purchase history can be generated automatically. These reports would help pharmacy owners analyze business performance, monitor transactions, and maintain proper records efficiently.

Future versions of the project may also include barcode scanning functionality for faster medicine billing and stock management. Barcode integration would reduce manual data

entry and improve billing accuracy. The system can also support online medicine ordering and home delivery services, which would make the application more modern and customer-friendly.

Cloud database integration is another important enhancement that can be added in the future. Cloud storage would help in maintaining secure backups of pharmacy records and allow access to data from multiple systems or locations. This would improve data safety, scalability, and accessibility.

Additional security features such as encrypted passwords, OTP verification, and role-based access control can also be implemented to improve privacy and system security. Different user roles such as admin, pharmacist, cashier, and manager can be created with separate permissions and access levels.

The user interface can also be improved further to provide a more interactive, responsive, and attractive experience for users. Mobile application support can also be added in future versions so that pharmacy operations can be managed through smartphones or tablets.

Artificial intelligence and data analytics features may also be integrated in the future for predicting medicine demand, analyzing sales trends, and recommending stock updates automatically. Such smart features can help pharmacies make better business decisions and improve operational efficiency.

Overall, the Pharmacy Management System has strong future scope and can be developed into a more advanced, intelligent, and fully-featured pharmacy management solution with the integration of modern technologies and advanced functionalities.

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